Iterative Spectral–Spatial Hyperspectral Anomaly Detection

Chein-I Chang, *Life Fellow, IEEE*, Chien-Yu Lin, Pau-Choo Chung, *Fellow, IEEE*, and Peter Fuming Hu

***Abstract*— Anomaly detection (AD) requires spectral and spatial information to differentiate anomalies from their surrounding data samples. To capture spatial information, a general approach is to utilize local windows in various forms to adapt local characteristics of the background (BKG) from which unknown anomalies can be detected. This article develops a new approach, called iterative spectral–spatial hyperspectral AD (ISSHAD), which can improve an anomaly detector in its performance via an iterative process. Its key idea is to include an iterative process that captures spectral and spatial information from AD maps (ADMaps) obtained in previous iterations and feeds these anomaly maps back to the current data cube to create a new data cube for the next iteration. To terminate the iterative process, a Tanimoto index (TI)-based automatic stopping rule is particularly designed. Three types of spectral and spatial information, ADMaps, foreground map (FGMap), and spatial filtered map (SFMap), are introduced to develop seven various versions of ISSHAD. To demonstrate its full utilization in improving AD performance, a large number of extensive experiments are performed for ISSHAD along with its detailed comprehensive analysis among several most recently developed anomaly detectors, including classic, dual-window-based, low-rank representation model-based, and tensor-based AD methods for validation.**

***Index Terms*— Anomaly detection (AD), iterative spectral–spatial anomaly detection (ISSHAD), Reed–Xiaoli anomaly detector (RXAD).**

NOMENCLATURE

AD	Anomaly detection.
ADP	Anomaly detection probability.
ADBS	Anomaly detection with BKG suppression.
AUC	Area under an ROC curve.
BKG	Background.
BDP	Background detection probability.
BS	Background suppression.
CDA	Component decomposition analysis.
CRD	Collaborative representation-based detector.
EAS	Effective anomaly space.
EPF	Edge preserving filter.
FG	Foreground.
GoDec	Go decomposition.
HSI	Hyperspectral imaging.
ISS	Iterative spectral–spatial.
ISSHAD	Iterative spectral–spatial hyperspectral anomaly detection.
ISSHAD-1.1	ISSHAD with feedback of AD maps.
ISSHAD-1.2	ISSHAD with feedback of foreground AD maps.
ISSHAD-1.3	ISSHAD with feedback of spatial-filtered AD maps.
ISSHAD-2.1	ISSHAD with feedback of ADMap∪FGMap.
ISSHAD-2.2	ISSHAD with feedback of ADMap∪SFMap.
ISSHAD-2.3	ISSHAD with feedback of FGMap∪SFMap.
ISSHAD-3	ISSHAD with feedback of ADMap∪FGMap∪SFMap.
FG	Foreground.
IC	Independent component.
JAD	Joint anomaly detectability.
JBS	Joint BKG suppression.
LRaSMD	Low-rank and sparse matrix decomposition.
LRR	Low-rank representation.
LSDM-MoG	Low-rank and sparse decomposition model with a mixture of Gaussians.
OADP	Overall anomaly detection probability.
PTA	Prior-based tensor approximation.
RXAD	Reed–Xiaoli anomaly detector.
ROC	Receiver operating characteristic.
SC	Sparsity cardinality.
SF	Spatial filter.
SNPR	Signal-to-noise probability ratio.
TI	Tanimoto index.

Manuscript received 12 August 2022; revised 17 January 2023; accepted 17 February 2023. Date of publication 22 February 2023; date of current version 3 March 2023. This work was supported in part by the National Science and Technology Council (NSTC) under Grant 111-2634-F-006-012. *(Corresponding author: Chien-Yu Lin.)*

Chein-I Chang and Chien-Yu Lin are with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County (UMBC), Baltimore, MD 21250 USA (e-mail: cchang@umbc.edu; chienyu2@umbc.edu).

Pau-Choo Chung is with the Department of Electrical Engineering, National Cheng Kung University, Tainan 70101, Taiwan (e-mail: pcchung@ee.ncku.edu.tw).

Peter Fuming Hu is with the R. A. Cowley Shock Trauma Center, Shock Trauma Anesthesia Organized Research Center, University of Maryland School of Medicine, Baltimore, MD 21201 USA (e-mail: phu@som.umaryland.edu).

Digital Object Identifier 10.1109/TGRS.2023.3247660

I. INTRODUCTION

DUE to significantly improved spectral resolution, HSI can reveal and uncover many unknown material substances, which cannot be identified by prior knowledge or visual inspection for data exploitation [1], [2]. Therefore, AD has been one of the major strengths that can be offered by HSI [3]. In particular, over the past years, many AD approaches have been developed. Among them is the one developed by Reed and Yu [4], referred to as RXAD that

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

has been one of the most widely used anomaly detectors and also has been used as a benchmark detector to extend to many of its variants [5], [6], [7], [8], [9], [10]. However, such extensions are somewhat limited due to the constraint of using a generalized likelihood ratio test (GLRT) to derive RXAD by Reed and Yu [4] and Robey et al. [5]. Most recently, Chang [11], [12] deviated from this GLRT-based approach to develop a signal-to-noise ratio (SNR)-based AD, which included GLRT-based anomaly detectors as its special cases. Subsequently, Chang [13] also extended the well-known constrained energy minimization (CEM) subpixel detector to the CEM anomaly detector (CEM-AD), which also included the correlation matrix **R**-based anomaly detector (R-AD) developed by Chang and Chiang [14] as its special case.

Since RXAD-based extension seems to reach its limit, this article reinvents the wheel to take a rather different approach, called ISSHAD, which is inspired by the iterative training sampling developed by Chang et al. [15], ISS classification by Zhong et al. [16], [17], and band sampling by Chang [18] and Chang and Ma [19]. Its idea is similar to the spectral–spatial feedback close network system proposed by Zhong et al. [17]. It takes advantage of spectral and spatial information provided by AD maps (ADMaps) and then feeds these pieces of information back to the current data cube to create a new expanded data cube, which will be further reimplemented by the same anomaly detector. The same process is repeated again in an iterative process. There are two key issues needed to be solved to make ISSHAD work. One is how to obtain spectral and spatial information from ADMaps for feedback. The other is how to terminate the iterative feedback process.

In ISS classification, the feedback of data samples classified into different classes provides all necessary classification information to improve and enhance classification accuracy throughout an iterative process. However, because of the unavailability of target knowledge, no such feedback can be used. Furthermore, unlike classification that can produce one classification map for one class, an anomaly detector can only produce one single ADMap, which may not be able to provide sufficient and necessary spectral and spatial information. Moreover, the maximum a posteriori (MAP)-based stopping rule used by ISS classification is not applicable to AD. Accordingly, a direct application of ISS classification developed in [16] and [17] is not feasible.

To address the feedback issue, we particularly introduce two types of information from an ADMap. One is spectral information, called FG map (FGMap) obtained from the binary map (BMap) resulting from applying Otsu's method [20] to threshold an ADMap, where FGMap retains the original ADMap values in the FG, while setting zeros to data samples in the BKG. The other is spatial information, referred to as a spatial filtered map (SFMap) obtained from applying an SF to the ADMap. As a result, a total of three types of spectral and spatial information can be generated from anomaly maps, ADmap, spectral FGMap, and spatial SFMap, all of which can be used for feedback information. Depending upon how to best utilize these pieces of information, seven variants of ISSHAD can be derived.

1) There are three feedbacks of one single anomaly map, i.e., ISSHAD-1.1 that feeds back only ADMap, ISSHAD-1.2 that feeds back only FGMap, and ISSHAD-1.3 that feeds back only SFMap.
2) There are three feedbacks of two single anomaly maps, i.e., ISSHAD-2.1 that feeds back ADMap and FGmap, ISSHAD-2.2 that feeds back ADmap and FGMap, and ISSHAD-2.3 that feeds back FGMap and SFMap.
3) There is only one feedback of three single anomaly maps, i.e., ISSHAD-3 that feeds back ADMap, FGMap, and SFMap.

To address the stopping rule issue, the MAP used in [16] and [17] is replaced by TI in [21], which is calculated based on Otsu's thresholded binary ADmaps in two consecutive iterations. If the value of TI is greater than a prescribed error threshold, the iterative process is then terminated.

Implementing the above seven different combinations of feedback along with the TI-based stopping rule results in seven ISSHAD variants. It should be noted that there is no particular anomaly detector specified in ISSHAD. This is because ISSHAD can be applied to any anomaly detector to further improve its detection performance. If the selected anomaly detector works poorly, ISSHAD can significantly improve its detection performance. On the other hand, if the selected anomaly detector performs very well, ISSHAD may not be able to improve its performance greatly but can still manage to improve its performance slightly to some extent.

Finally, in order to validate the utilization of ISSHAD, several state-of-the-art AD methods recently developed in the literature were selected for experiments to represent different categories of AD methods, such as the dual-window-based CRD by Li and Du [22], the low rank and sparse representation-based LRR by Xu et al. [23], the BKG estimated-based LSDM-MoG by Li et al. [24], the tensor decomposition-based PTA by Li et al. [25], CDA by Chen et al. [26], and EAS by Chang [27]. Since each of these test anomaly detectors has its own strengths and weaknesses, the traditional 2-D ROC analysis solely based on the curves of detection probability, P_D versus false alarm probability, P_F, is not effective to be used for performance evaluation. Fortunately, Chang [28] recently developed a 3-D ROC curve-based analysis that can be used to resolve this issue where eight 3-D ROC curve-derived detection measures enable one to evaluate an anomaly detector in all round perspectives. As a result, a very large body of experiments constitutes a crucial part of this article to provide hard evidence that ISSHAD is indeed a versatile AD technique that can improve most of the currently used AD methods in detection performance.

The main contributions can be summarized as follows.

1) Three types of spectral- and spatial-captured maps can be generated from an anomaly map, that is, ADMap, spectral FGMap, and spatial SFMap for feedback information.
2) Seven versions can be derived for ISSHAD using different combinations of spectral- and spatial-captured maps.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

3) Extensive experiments and detailed discussions are included to demonstrate the advantages of ISSHAD. Despite that these experiments and discussions take up most of the space, such comprehensive study and analysis are vital parts to justify the utilization of ISSHAD. In particular, the experiments further demonstrate that the commonly used the San Diego airport scene is not suitable for AD. As a result, many reports based on this scene and its similar data scenes generally mislead to unreliable conclusions.

The remainder of this article is organized as follows. Section II briefly reviews related works to the most recently developed AD methods. Section III describes key ideas in the design of ISSHAD and further develops seven variants of ISSHAD. Section IV introduces 3-D ROC curve-derived detection measures. Section V describes the three hyperspectral images widely used for AD experiments. Section VI presents and discusses experimental results. Section VII draws conclusions.

II. Related Works

Many works on AD have been reported in the literature. One most widely used approach is RXAD, also known as the Mahalanobis distance, which used the global sample autocovariance matrix $\mathbf{K}$ to perform AD. It was later extended to an anomaly detector developed in [14], referred to as R-AD in [3], which replaced $\mathbf{K}$ with the global sample autocorrelation matrix $\mathbf{R}$. The details of theoretical derivations of RXAD and R-AD from an HSI perspective can be found in [12]. Since RXAD and R-AD were introduced, many RXAD/R-AD-type detectors and variants have been developed in the past. Examples include local RXAD [29], [30] and support vector data description (SVDD) by Banerjee et al. [31], weighted-RXAD and linear filter-based RXAD by Guo et al. [32], a sliding window by Chang et al. [33], a dual window-based eigen separation transform (DWEST) by Kwon et al. [34] followed by CRD, which also used dual windows in [22], multiple-window AD (MWAD) by Liu and Chang [35] by making use of multiple windows to perform AD adaptively, guided filtering-based AD in [36], spectral–spatial feature extraction-based AD by Lei et al. [37], and a nonlinear anomaly detector, kernel-based RXAD by Kwon and Nasrabadi [38]. In addition, there were also AD methods developed for various applications, such as unlabeled classification by Wang et al. [39] and real-time processing of AD by Chen et al. [40]. Recently, Liu et al. [41] extended AD by considering the Gaussian BKG with an unknown covariance matrix from which two equivalent adaptive GLRT-based detectors were derived. Most interestingly, Chang [11] showed that GLRT can be indeed obtained without using the Gaussian assumption. Yuan et al. [42] proposed an AD method by constructing vertex- and edge-weighted graphs to detect anomalies with better accurate detection performance and better robustness to noise where no BKG distributions are required. Li et al. [43] developed an attribute- and edge-preserving filtering-based detection (AED) method by fusing the multiscale information extracted by attribute- and edge-preserving filters. However, as pointed out by Chang [12], a crucial difference between target detection and AD is that the former is active target detection, which assumes that the targets to be detected must be known, while the latter is passive target detection, which does not require any target knowledge at all. This taxonomy was initially used by military intelligence applications where active target detection is considered reconnaissance, which requires the prior target knowledge to be used for target detection as opposed to passive target detection considered as surveillance without knowing any target knowledge. As a result, active target detection is generally evaluated by the detectability of known targets to be detected. This is why binary simple hypothesis testing has been used to detect targets in noise. By contrast, since AD does not know the targets it detects, its performance must rely on its ability in BS. By means of BS, the contrast enhancement between BKG and anomalies can bring targets of interest out of BKG to light. Thus, two key features that have significant impacts on AD are BS and the detection enhancement of anomalies. These two issues have triggered the most recent trends in developing and designing AD algorithms.

As for the BS issue, one of the most recent efforts is low rank and sparse representation-based approaches, which considered BKG to be represented by a low-rank space. A typical example is [23] that developed an LRR method for AD based on the assumption that a hyperspectral dataset can be effectively decomposed into a low-rank component for BKG and a sparse component for anomalies via imposing a spatial constraint on the low-rank representation coefficients smoothed by single or multiple local window strategies. Two other approaches are also of interest. One is BKG estimation by using BKG training samples, such as deep learning networks by Ma et al. [44] and Wang et al. [45]. The other is BKG reconstruction by the autoencoder (AE) by Xie et al. [46] and the generative adversarial network (GAN) by Zhong et al. [47] and Jiang et al. [48]. In either case, the estimation/reconstruction error is assumed to be the main cause, which results in poor AD.

As for AD enhancement, Qu et al. [49] developed a Gaussian mixture model (GMM)-based approach to anomaly extraction and an effective GMM-based weighting approach for fusing the extracted anomaly maps, which can be further rectified by using a guided filter to obtain the final ADMap. Zhao et al. [50] developed a robust nonlinear AD (RNAD) that resolved the corruption issue resulting from anomalous targets via nonlinearly mapping the input data into high-dimensional kernel feature space. Later, Zhao et al. [51] also proposed a sparsity score estimation framework for HAD to separate BKG from anomalies using an overcomplete dictionary and its corresponding sparse code matrix. Recently, Chang et al. [52] further developed a BKG-anomaly component projection and separation optimized filter (BASO) to separate anomalies from BKG.

The main focus of the abovementioned approaches is centered on either BKG estimation/reconstruction or AD enhancement. A more effective means is to jointly consider BKG and AD enhancement issues together. In particular, LRaSMD model-based methods have received considerable interest and been developed to separate BKG and anomalies into two different separate spaces. Several specific models are of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

particular interest. One is robust principal component analysis (RPCA) developed by Candes et al. [53] along with its variant, which is an approach, called robust subspace learning (RSL) developed by Vaswani et al. [54], [55], Bouwmans et al. [56], and Sobral et al. [57], via sparse + low-rank (S + LR) matrix decomposition. Another is an approach, called GoDec developed by Zhou and Tao [58], which represents a data matrix $\mathbf{X}$ as a linear decomposition of a low-rank matrix, $\mathbf{L}$, and a sparse matrix, $\mathbf{S}$, plus a noise matrix $\mathbf{N}$ by $\mathbf{X} = \mathbf{L}+\mathbf{S}+\mathbf{N}$. As an alternative model to LRaSMD, CDA was recently developed by Chen et al. [26], which decomposes a data matrix into the principal component (PC) and IC spaces.

However, the abovementioned models only offer separation of these two spaces with anomalies still sandwiched between BKG and noise. Thus, in order to perform AD effectively, BKG and anomalies must be addressed simultaneously. For example, Wang et al. [59] proposed an autonomous hyperspectral AD (Auto-AD) network to reconstruct the BKG by including an adaptive-weighted loss function to assign greater weights to potential anomalous pixels, so as to increase the contrast of the anomalies against BKG. Li et al. [24] developed an LSDM-MoG where the sparse component can be described by a mixture of Gaussian noises so that anomalies can be detected by the Manhattan distance in [60], which took advantage of the LRaSMD model along with RXAD to develop an AD method. Li et al. [25] developed a PTA to decompose the data space into a BKG tensor and an anomaly tensor for a hyperspectral AD so that the anomalies can be detected in the anomaly tensor. Most recently, Chang [27] constructs an EAS for HAD, which makes use of data sphering (DS) and IC analysis (ICA) techniques to take care of both BKG and noise altogether in a single space. It turns out that EAS becomes one of the best AD methods compared to recently developed AD methods as will be demonstrated in the following experiments.

III. ISSHAD

All the AD methods reviewed in Section II are specifically designed to achieve the best possible detection performance in one way or another. To improve these methods, their original versions must be modified or revised. It will be difficult for some AD methods, which already perform very well, such as CRD in [22]. To the best of our knowledge, there exists no work to improve the detection performance of an AD method without revising its architecture and parameters. Interestingly, ISSHAD can do just that including CRD.

A. Motivation of ISSHAD

The development of ISSHAD originated from the nature of anomalies that require spatial correlation among the spectral characteristics of their surrounding pixels. Many AD methods have been developed to address this issue. ISSHAD deviates from the traditional wisdom of using windows, deep learning networks, and low rank and sparse representation model-based approaches. Instead, ISSHAD is designed to improve and enhance the detection performance of an AD method, which can be any existing AD method including those that have been shown to perform very well.

ISSHAD is quite different from all the reported works in the literature. Its idea is inspired by a CEM approach to hyperspectral image classification, referred to as iterative CEM (ICEM)-based classification by Kang et al. [61]. Since CEM is a subpixel detection technique, ICEM treats each class as a desired target, utilizes a set of SFs to iteratively capture spatial information from CEM-detected maps, and feeds these spatial-filtered detection maps back to create new data cubes from the previous data cubes for follow-up iterations. This iterative feedback process allows CEM to include more spatial information, so as to improve classification accuracy iteration-by-iteration. Unfortunately, such a CEM-based approach seems not directly applicable to AD because CEM requires prior knowledge of desired targets, i.e., class information [12], [13]. The ISSHAD presented in this article resolves this issue by introducing an ISS feedback process into an AD method that can iteratively update spatial information from its detection maps via expanded data cubes obtained by including feedback of spatial filtered detection maps through iterations. During the entire iterative process, the same AD method is implemented repeatedly without changing its architecture. Most importantly, the anomaly maps used for feedback are actually obtained directly from the same AD method. Consequently, ISSHAD is immune from any specific AD method and can be applied to any arbitrary AD method.

The key to making ISSHAD work is how to obtain feedback information from a used AD method. Thus, first, we need to select an AD method needed to be improved and also an SF, generally Gaussian filter or EPF used in [62] to capture spatial information. Then, there are three types of spatial information that can be obtained from the selected AD method and SF. The first one is the anomaly map produced by the used AD method, denoted by ADMap, which is real-valued. The second one is the FGMap that is obtained by retaining the original real values of FG pixels while setting all the BKG samples to zeros where the FG and BKG are obtained by Otsu's method. The third one is spatial filtered anomaly maps obtained by applying the SF to the ADMap. Depending upon how these three types of anomaly maps are used as feedback information, seven versions of ISSHAD can be derived.

1) *One Single Feedback Anomaly Map:* ISSHAD-ADMap, ISSHAD-FGMap, and ISSHAD-SFMap.
2) *Two Feedback Anomaly Maps:* ISSHAD-ADMap ∪FGMap (ISSHAD-2.1), ISSHAD-ADMap∪ SFMap (ISSHAD-2.2), and ISSHAD-FGMap∪SFMap (ISSHAD-2.3).
3) *Three Feedback Anomaly Maps:* ISSHAD-ADMap∪FGMap∪SFMap (ISSHAD-3) where A, A∪B, and A∪B∪C indicate that A, both A and B, and all the three A, B, and C are fed back to the current data cube to create a new data cube. The same feedback loop is implemented iteration-by-iteration until it reaches a stopping rule, which can be determined by TI.

B. Design of ISSHAD

The design of ISSHAD consists of the following major components.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

1) Selection of an AD method to be implemented in ISSHAD for improvement.
2) Selection of an SF used for ISSHAD. In this article, EPF [62] was used.
3) Feedback of three types of anomaly map information to ISSHAD.
   a) *ADMap:* Detection map produced by the used anomaly detector.
   b) *FGMap:* FG anomaly map obtained from the BMap by Otsu's method

$$\text{FGMap}^{(j)} = \begin{cases} \text{FGMap}^{(j)}_{m,n} = 0, \\ \quad \text{if } \text{BMap}^{(j-1)}_{m,n} = 0 \\ \text{FGMap}^{(j)}_{m,n} = \text{ADMap}^{(j)}_{m,n}, \\ \quad \text{if } \text{BMap}^{(j-1)}_{m,n} = 1. \end{cases} \tag{1}$$

   c) *SFMap:* Spatial filtered anomaly map obtained by applying an SF to the ADMap.

   It is worth noting that the feedback of AD information in this step is key to deriving the seven variants of ISSHAD, presented in Section III-C.
4) TI-based stopping rule for ISSHAD

$$\text{TI}^{(j)} = \frac{\left|\text{BMap}^{(j)} \cap \text{BMap}^{(j-1)}\right|}{\left|\text{BMap}^{(j)} \cup \text{BMap}^{(j-1)}\right|} > \varepsilon. \tag{2}$$

C. Seven Variants of ISSHAD

According to step 3 described in ISSHAD, the feedback of different combinations of the spectral and spatial AD information captured from anomaly maps yields seven variants of ISSHAD.

1) Feedback of a Single Anomaly Map: There are three ways to feed one single anomaly map back to the current data cube.

(1.1) ISSHAD-ADMap (ISSHAD-1.1) feeds the ADMap back to create a new data cube, $\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{ADMap}^{(j)}$. Its algorithm of step-by-step implementation is given as follows.

ISSHAD-ADMap (ISSHAD-1.1)

1. Initial condition:
   Select an anomaly detector and a spatial filter (SF)
   Set $\Omega^{(j)} = \text{HSI}$, a threshold ε and an iteration count $j = 0$
2. At the jth iteration
   Implement the selected anomaly detector on $\Omega^{(j)}$ to produce $\text{ADMap}^{(j)}$.
3. Use Otsu's method to threshold $\text{ADMap}^{(j)}$ to produce $\text{BMap}^{(j)}$.
4. Check if $j = 0$, then $j \leftarrow j+1$ and go to step 2. Otherwise, continue.
5. Check if

$$\text{TI}^{(j)} = \frac{\left|\text{BMap}^{(j)} \cap \text{BMap}^{(j-1)}\right|}{\left|\text{BMap}^{(j)} \cup \text{BMap}^{(j-1)}\right|} > \varepsilon \tag{3}$$

   then the iterative process is terminated. Otherwise, continue.
6. Let $j \leftarrow j+1$ and expand the current data cube by

$$\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{ADMap}^{(j)} \tag{4}$$

   Go to step 2.

(1.2) ISSHAD-FGMap (ISSHAD-1.2) feeds the FGMap back to create a new data cube $\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{FGMap}^{(j)}$. Its algorithm of step-by-step implementation is given as follows.

ISSHAD-FGMap (ISSHAD-1.2)

1. Steps 1-5 are identical to ISSHAD-ADMap
6. Create a foreground map $\text{FGMap}^{(j)}$ by

$$\text{FGMap}^{(j)} = \begin{cases} \text{FGMap}^{(j)}_{m,n} = 0 \text{ if } \text{BMap}^{(j-1)}_{m,n} = 0 \\ \text{FGMap}^{(j)}_{m,n} = \text{ADMap}^{(j-1)}_{m,n} \text{ if } \text{BMap}^{(j-1)}_{m,n} = 1 \end{cases} \tag{5}$$

7. Let $j \leftarrow j+1$ and expand the current data cube by $\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{FGMap}^{(j)}$.
   Go to step 2.

(1.3) ISSHAD-SFMap (ISSHAD-1.3) feeds the spatial filtered anomaly map obtained from applying an SF to ADMap back to create a new data cube, $\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{SFMap}^{(j)}$. Its algorithm of step-by-step implementation is given as follows.

ISSHAD-SFMap (ISSHAD-1.3)

1. Steps 1-5 are identical to steps 1-5 in ISSHAD-ADMap.
6. Apply the selected SF to $\text{ADMap}^{(j)}$ to produce $\text{SFMap}^{(j)}$.
7. Let $j \leftarrow j+1$ and expand the current data cube,

$$\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{SFMap}^{(j)} \tag{6}$$

   Go to step 2.

2) Feedback of Two Anomaly Maps:

(2.1) ISSHAD-2.1 feeds ADMap and FGMap back to the current data cube to create a new data cube, $\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{ADMap}^{(j)} \cup \text{FGMap}^{(j)}$. Its algorithm of step-by-step implementation is given as follows.

ISSHAD-2.1

1. Steps 1-5 are identical to steps 1-5 in ISSHAD-ADMap.
6. Find $\text{FGMap}^{(j)}$ by ().
7. Let $j \leftarrow j+1$ and expand the current data cube,

$$\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{ADMap}^{(j)} \cup \text{FGMap}^{(j)} \tag{7}$$

   Go to step 2.

(2.2) ISSHAD-2.2 feeds ADMap and SFMap back to the current data cube to create a new data cube, $\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{ADMap}^{(j)} \cup \text{SFMap}^{(j)}$. Its algorithm of step-by-step implementation is given as follows.

ISSHAD-2.2

1. Steps 1-5 are identical to steps 1-5 in ISSHAD-ADMap.
6. Apply the selected SF to $\text{ADMap}^{(j)}$ to produce $\text{SFMap}^{(j)}$.
7. Let $j \leftarrow j+1$ and expand the current data cube,

$$\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{ADMap}^{(j)} \cup \text{SFMap}^{(j)} \tag{8}$$

   Go to step 2.

(2.3) ISSHAD-2.3 feeds FGMap and SFMap back to the current data cube to create a new data cube, $\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{FGMap}^{(j)} \cup \text{SFMap}^{(j)}$. Its algorithm of step-by-step implementation is given as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

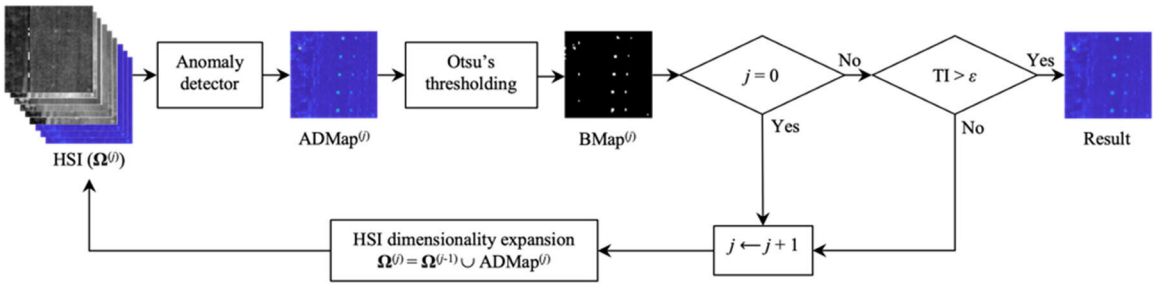


Fig. 1. Graphic diagram of implementing the jth iteration of ISSHAD-1.1.

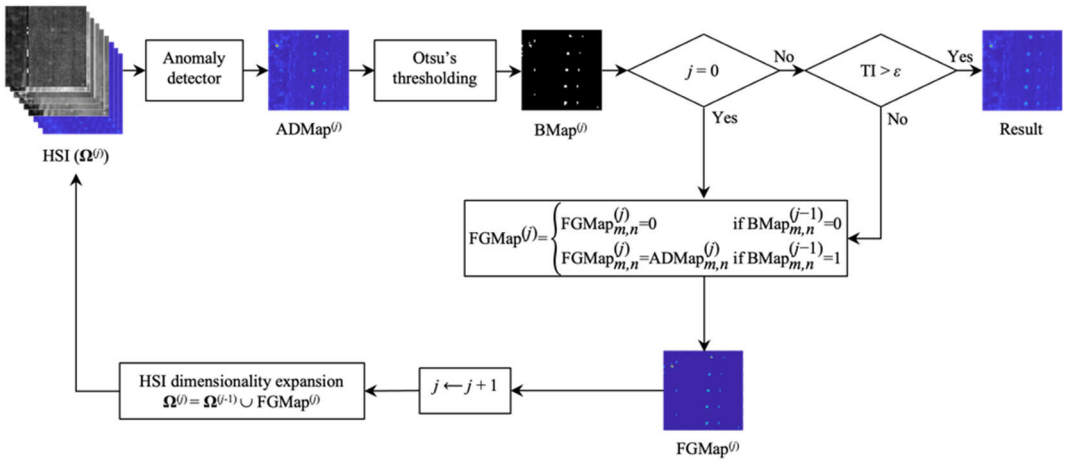


Fig. 2. Graphic diagram of implementing the jth iteration of ISSHAD-1.2.

ISSHAD-2.3

1. Steps 1-5 are identical to ISSHAD-ADMap
6. Find FGMap$^{(j)}$ by ().
7. Apply the selected SF to ADMap$^{(j)}$ to produce SFMap$^{(j)}$.
8. Let $j \leftarrow j+1$ and expand the current data cube by

$$\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{FGMap}^{(j)} \cup \text{SFMap}^{(j)}. \tag{9}$$

Go to step 2.

3) Feedback of Three Anomaly Maps: ISSHAD-3 feeds ADMap, FGMap, and SFMap back to the current data cube to create a new data cube, $\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{SFMap}^{(j)} \cup \text{FGMap}^{(j)} \cup \text{ADMap}^{(j)}$.

ISSHAD-3

1. Steps 1-5 are identical to ISSHAD-ADMap
6. Find FGMap$^{(j)}$ by ().
7. Apply the selected SF to ADMap$^{(j)}$ to produce SFMap$^{(j)}$.
8. Let $j \leftarrow j+1$ and expand the current data cube by

$$\mathbf{\Omega}^{(j)} = \mathbf{\Omega}^{(j-1)} \cup \text{SFMap}^{(j)} \cup \text{FGMap}^{(j)} \cup \text{ADMap}^{(j)} \tag{10}$$

Go to step 2.

Figs. 1–7 depict the flow diagrams of implementing the jth iteration of the above seven ISSHAD variants.

In order to demonstrate and illustrate the great potential and benefits provided by ISSHAD that does not require any specific AD method, extensive experiments and comprehensive analysis were conducted based on seven AD methods in different categories: classic RXAD, dual window-based CRD, low rank and sparse representation-based LRR, Gaussian mixture-estimated BKG-based LSDM-MoG, tensor decomposition-based PTA, component-based CDA, and EAS, which spheres data and then uses ATGP-ICA [27] to produce the first j ICs to form an IC component subspace from which an EAS can be constructed by imposing SC in [58] on the IC component space. The experimental results show that ISSHAD can significantly improve the performance of all compared AD methods with the exception that CDA has instability in feedback.

IV. 3-D ROC Curve-Derived Detection Measures

Due to the unavailability of prior target knowledge, AD performance is significantly affected by BKG. The standard 2-D ROC analysis can only evaluate the effectiveness of a detector but cannot address the issue of anomaly detectability, BKG suppressibility, and JAD with BKG suppressibility. Since ISSHAD is particularly designed to improve an AD method in all these aspects, using 2-D ROC curves clearly cannot meet this need. Therefore, the 3-D ROC curves derived in [28] can be used to serve this purpose. It includes τ, which is the threshold parameter used to determine if a data sample is an anomalous sample, as a third parameter to plot a curve of three parameters, (P_D,P_F, τ). By virtue of this 3-D ROC curve, three 2-D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) can be generated to calculate the AUC, denoted by AUC$_{(D,F)}$, AUC$_{(D,\tau)}$, and AUC$_{(F,\tau)}$, respectively, from which eight derived detection measures can be derived as follows.

A. Effectiveness of a Detector

As noted in [28], the standard 2-D ROC curve of (P_D, P_F) is generated by the same threshold τ, and P_D and P_F

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

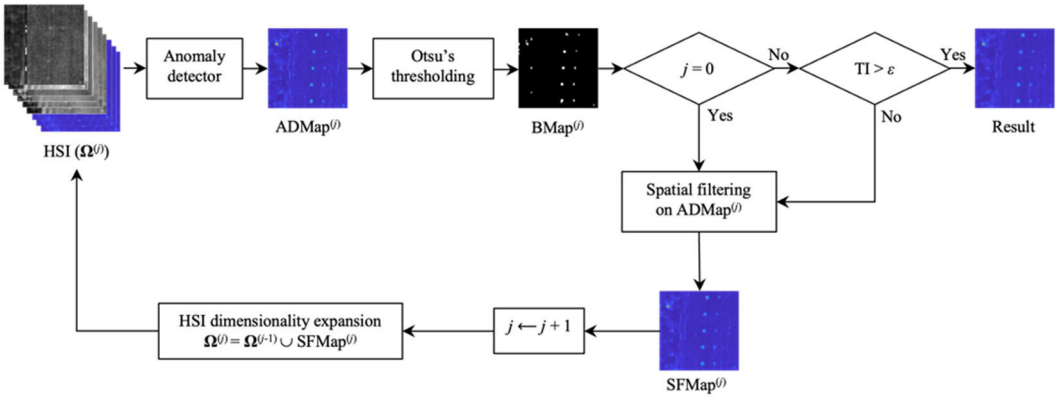


Fig. 3. Graphic diagram of the implementation of ISSHAD-1.3.

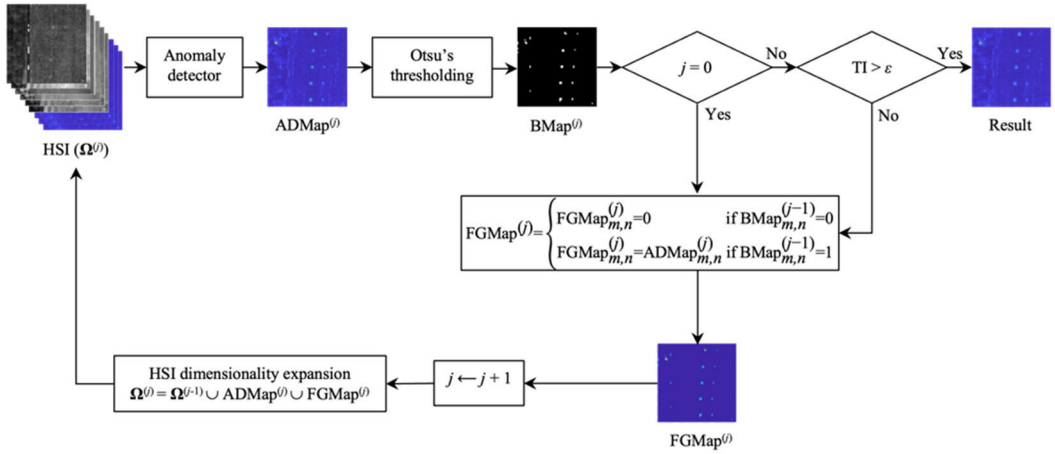


Fig. 4. Graphic diagram of t implementing the jth iteration of ISSHAD-2.1.

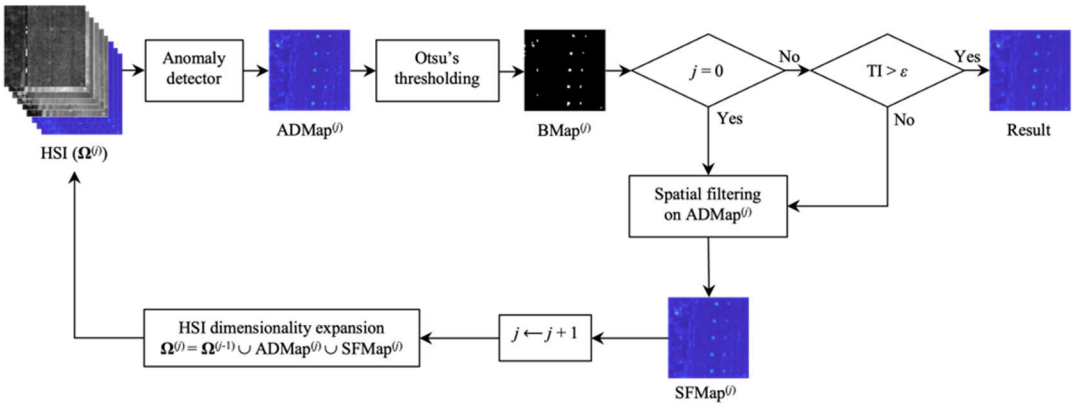


Fig. 5. Graphic diagram of implementing the jth iteration of ISSHAD-2.2.

cannot work alone. In other words, when P_D is high, so is P_F. Conversely, when P_D is low, so is P_F. As a result, $\text{AUC}_{(D,F)}$ can be only used to evaluate the effectiveness of a detector given by

$$0 \leq \text{AUC}_{(D,F)} \leq 1 \tag{11}$$

but cannot evaluate anomaly detectability, BKG suppressibility, and JAD with BKG suppressibility.

B. Anomaly Detection Probability

The 2-D ROC curve of (P_D, τ) is a plot of P_D versus τ. Thus, it can be used to evaluate anomaly detectability. That is, the higher is the value of $\text{AUC}_{(D,\tau)}$, the better is the anomaly detector. In this case, we can define a measure, called ADP of a detector as AUC_{ADP} by

$$0 \leq \text{AUC}_{\text{ADP}} \leq 1 \tag{12}$$

and JAD

$$0 \leq \text{AUC}_{\text{JAD}} = \text{AUC}_{(D,F)} + \text{AUC}_{\text{ADP}} \leq 2. \tag{13}$$

C. Background Detection Probability

In signal detection in noise, P_F is the false alarm probability, and $1 - P_F$ is the detection of noise. Thus, using $1 - P_F$ to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

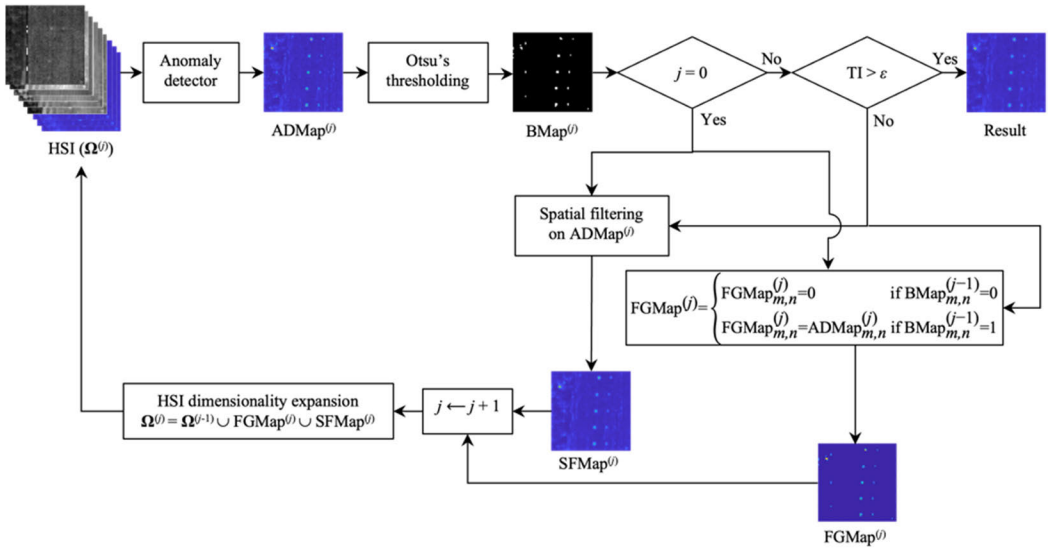


Fig. 6. Graphic diagram of implementing the jth iteration of ISSHAD-2.3.

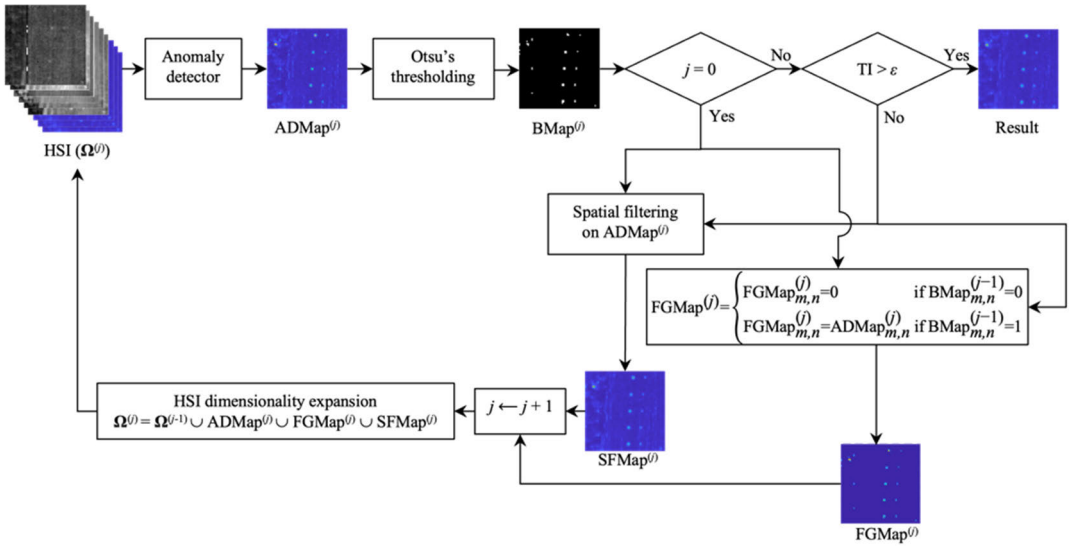


Fig. 7. Graphic diagram of implementing the jth iteration of ISSHAD-3.

evaluate noise detection performance for signal detection does not make sense. However, this is not true from AD, where $1 - P_F$ is actually the detection probability of BKG. Thus, we can define BDP, AUC_{BDP}, as a counterpart of ADP in (12) by

$$0 \leq \text{AUC}_{\text{BDP}} = 1 - \text{AUC}_{(F,\tau)} \leq 1 \tag{14}$$

and joint BKG suppressibility (JBS)

$$0 \leq \text{AUC}_{\text{JBS}} = \text{AUC}_{(D,F)} + \text{AUC}_{\text{BDP}} \leq 2. \tag{15}$$

D. Joint Anomaly Detectability With Background Suppressibility

As pointed out in Sections IV-A, P_D and P_F are calculated by the same threshold τ for their joint performance without accounting for ADP and BDP separately. In order to factor $\text{AUC}_{(F,\tau)}$ into $\text{AUC}_{(D,\tau)}$, we need to define JAD with the BKG suppressibility of a detector.

1) AD in BKG: The first measure is to consider AD and BS working separately with different values of the threshold τ as an independent parameter. Since P_F is caused by the probability of misdetecting BKG as anomalies, this error probability should be subtracted from ADP. To take care of this effect, a measure, factoring P_F into ADP, to be called ADBS, can be defined as AUC_{ADBS} by

$$-1 \leq \text{AUC}_{\text{ADBS}} = \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \leq 1. \tag{16}$$

2) Overall Anomaly Detection Probability: Another important measure is derived from a classification point of view. It is called OADP defined by

$$0 \leq \text{AUC}_{\text{OADP}} = \text{AUC}_{\text{ADP}} + \text{AUC}_{\text{BDP}} \leq 2 \tag{17}$$

with AUC_{ADP} in (12) and AUC_{BDP} in (14) with $0 \leq \rho \leq 1$ being the probability to weigh AUC_{ADP}.

3) Signal-to-Noise Probability Ratio: The last but most effective detection measure comes from an idea similar to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

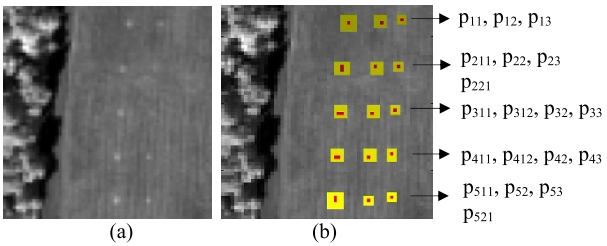


Fig. 8. (a) HYDICE15-panel scene. (b) Ground-truth map of 19 R panel pixels.



Fig. 9. HYDICE urban scene. (a) HYDICE urban. (b) Selected area. (c) Ground-truth map.



Fig. 10. San Diego Airport scene. (a) San Diego airport. (b) Selected area. (c) Ground-truth map.

SNR, called SNPR, defined by

$$0 \le \mathrm{AUC}_{\mathrm{SNPR}} = \frac{\mathrm{AUC}_{(D,\tau)}}{\mathrm{AUC}_{(F,\tau)}} \tag{18}$$

where the false alarm probability P_F can be assumed to be caused by noise corresponding to BKG.

V. Images Used for Experiments

A. HYDICE 15-Panel Scene

An airborne hyperspectral digital imagery collection experiment (HYDICE) scene shown in Fig. 8 was collected in August 1995 from a flight altitude of 10 000 ft with a ground sampling distance of approximately 1.56 m. This scene has been studied extensively by many reports, such as [1], [2], and [38]. It has a total of 169 bands that were used for the experiments with low signal/high noise bands: bands 1–3 and bands 202–210, and water vapor absorption bands: bands 101–112 and bands 137–153 were removed. There are 15 square panels in Fig. 8(a) with three different sizes, 3 m × 3 m, 2 m × 2 m, and 1 m × 1 m, respectively. Due to the ground sampling distance of approximately 1.56 m, each of the panels in the first column except the first row contains two panel pixels highlighted by red, p_{211} and p_{221} in row 2, p_{311} and p_{312} in row 3, p_{411} and p_{412} in row 4, and p_{511} and p_{521} in row 5, as shown in Fig. 4. All the remaining 11 panels in Fig. 8 contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, and p_{13} in row 1, p_{22} and p_{23} in row 2, p_{32} and p_{33} in row 3, p_{42} and p_{43} in row 4, and p_{52} and p_{53} in row 5. Therefore, there are a total of 19 red (R) panel pixels. Fig. 8(b) shows their precise spatial locations with the pixels in yellow (Y pixels), indicating panel pixels mixed with the BKG.

TABLE I
Parameters Used for Test AD Methods

Anomaly Detector	Parameter setting
Spatial filter	EPF-G-g in [62]
CRD	$(w_{out},w_{in}) = (11,9)$
LRR	$\beta = 0.1$
LDSM-MoG	$(l_0,K) = (20,2)$ $(t_0,\mu_0,a_0,b_0,c_0,d_0,\alpha_0,\beta_0) = (10^{-3},0,10^{-6},10^{-6},10^{-6},10^{-6},10^{-6},10^{-6})$
PTA	$(\gamma,\alpha,\mu,\beta,\tau,\mathrm{maxIter}) = (1,1.7,0.069,0.1,400)$
CDA (panel)	$(p,m,j) = (13,7,6)$, detector $\delta^{\mathrm{RXAD}}_{\mathrm{PC}^7+\mathrm{IC}_6}$
CDA (urban)	$(p,\mathrm{m},j) = (9,5,4)$, detector $\delta^{\mathrm{RXAD}}_{\mathrm{PC}^5+\mathrm{IC}^4}$
CDA (San Diego)	$(p,m,j) = (10,2,8)$, detector $\delta^{\mathrm{RXAD}}_{\mathrm{PC}^2+\mathrm{IC}^8}$
ICSC (panel)	$(p,j) = (13,6)$, detector $\delta^{\mathrm{RXAD}}_{\mathrm{IC}^4}$
ICSC (urban)	$(p,j) = (9,4)$, detector $\delta^{\mathrm{RXAD}}_{\mathrm{IC}^6}$
ICSC (San Diego)	$(p,j) = (10,8)$, detector $\delta^{\mathrm{RXAD}}_{\mathrm{IC}^8}$
Max number of iterations	20 for LRR, LSDM-MoG, CDA

B. HYDICE Urban Scene

A second dataset is another HYDICE image with pseudocolor shown in Fig. 9(a) on the website at https://rslab.ut.ac.ir/data. It is an urban scene and comprised 210 spectral bands with 174 bands being used for experiments after the noise and water absorption bands had been removed. A region with a size of 80 × 100 pixels located at the upper right of the scene was selected as the test image shown in Fig. 9(b) along with the ground-truth map shown in Fig. 9(c) where 21 pixels were identified as anomalies, which were cars and roofs, because they had spectra that differ from BKG.

C. San Diego Airport Data

A third dataset used for experiments is an airborne visible/infrared imaging spectrometer (AVIRIS) acquired scene, which is a San Diego airport scene in CA, USA, with pseudocolor shown in Fig. 10(a). It is an urban scene in which the main BKG materials are roof, shadow, and grass. It has a size of 400 × 400 pixels with a 3.5-m spatial resolution and 224 spectral channels in wavelengths ranging from 370 to 2510 nm. After removing the bands that correspond to the water absorption regions, low SNR, and bad bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 available bands of the data were retained in the experiments. An area of 100 × 100 pixels at the upper left corner of the scene was selected as the test image shown in Fig. 10(b) along with its ground-truth map shown in Fig. 10(c).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

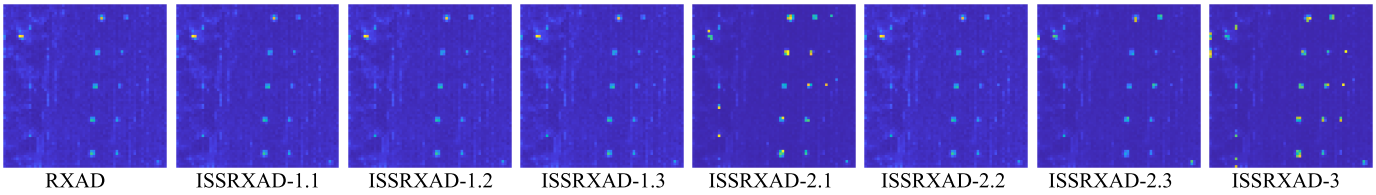


Fig. 11. ADMaps produced by RXAD along with its seven ISSRXAD variants on the HYDICE Panel scene.

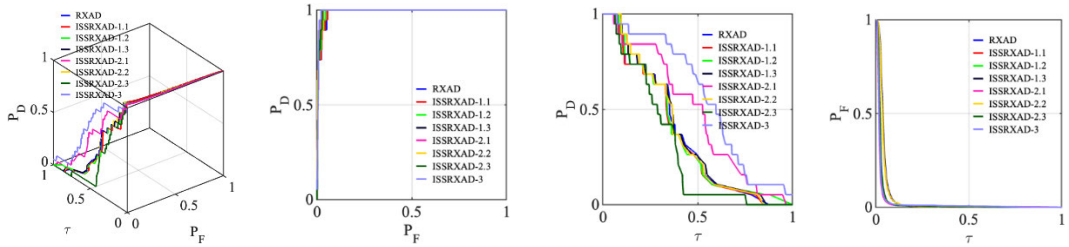


Fig. 12. 3-D ROC curves and the corresponding 2-D ROC curves produced by RXAD along with its seven ISSRXAD variants on the HYDICE Panel scene.

TABLE II
AUC VALUES PRODUCED BY RXAD ALONG WITH ITS SEVEN ISSRXAD VARIANTS ON THE HYDICE PANEL SCENE

Anomaly Detector	j	$\mathrm{TI}_{j,j-1}$	$\mathrm{AUC}_{(D,F)}$	$\mathrm{AUC}_{\mathrm{ADP}}$	$\mathrm{AUC}_{\mathrm{BDP}}$	$\mathrm{AUC}_{\mathrm{JAD}}$	$\mathrm{AUC}_{\mathrm{JBS}}$	$\mathrm{AUC}_{\mathrm{ADBS}}$	$\mathrm{AUC}_{\mathrm{SNPR}}$	$\mathrm{AUC}_{\mathrm{OADP}}$	Time(s)
RXAD	0	0.000	0.99	0.3574	0.9573	1.3474	1.9473	0.3147	8.3663	1.3147	0.157
ISSRXAD-1.1	1	0.976	0.9902	0.348	0.9596	1.3382	1.9498	0.3075	8.6098	1.3075	2.184
ISSRXAD-1.2	1	0.954	0.9918	0.3581	0.9589	1.3499	1.9507	0.317	8.7065	1.317	2.131
ISSRXAD-1.3	1	0.976	0.9933	0.3679	0.9578	1.3612	1.9512	0.3257	8.7244	1.3257	2.101
ISSRXAD-2.1	14	0.961	0.9933	0.4741	**0.9762**	1.4674	**1.9695**	0.4503	19.9084	1.4503	16.579
ISSRXAD-2.2	1	0.976	**0.9936**	0.3628	0.9592	1.3564	1.9528	0.3221	8.9004	1.3221	2.087
ISSRXAD-2.3	5	0.958	0.9919	0.2919	0.9726	1.2838	1.9645	0.2646	10.6573	1.2646	6.963
ISSRXAD-3	11	0.966	**0.9936**	**0.5402**	0.9739	**1.5338**	1.9675	**0.5141**	**20.6993**	**1.5141**	13.254

VI. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents a very detailed comprehensive study of experiments and a comparative analysis among seven test AD methods: RXAD, CRD, LRR, LSDM-MoG, PTA, CDA, and IC sparsity cardinality (ICSC) on three real scenes described in Section V. Because each AD method has its own merit and each data scene has also its own image characteristics, a very large number of experiments were performed under MATLAB R2021b with macOS Monterey version 12.2, CPU chip Apple M1 Pro 8 cores, 16-GB memory.

Table I shows the parameter setting for the test AD methods compared in this article.

A. HYDICE Panel Scene

1) RXAD: Fig. 11 shows the ADMaps of RXAD along with its corresponding seven ISSRXAD variants, ISSRXAD-1.1, ISSRXAD-1.2, ISSRXAD-1.3, ISSRXAD-2.1, ISSRXAD-2.2, ISSRXAD-2.3, and ISSRXAD-3 for the HYDICE panel scene, where ISSRXAD-3 gave the best result by visual inspection because it detected subpixel panels in the third column. Nonetheless, the other six versions were also produced comparable or better results than that produced by RXD. To conduct quantitative analysis, we plotted their 3-D ROC curves along with their corresponding three 2-D ROC curves in Fig. 12 where eight detection measures, $\mathrm{AUC}_{(D,F)}$, $\mathrm{AUC}_{\mathrm{ADP}}$, $\mathrm{AUC}_{\mathrm{BDP}}$, $\mathrm{AUC}_{\mathrm{JAD}}$, $\mathrm{AUC}_{\mathrm{JBS}}$, $\mathrm{AUC}_{\mathrm{ADBS}}$, $\mathrm{AUC}_{\mathrm{SNPR}}$, and $\mathrm{AUC}_{\mathrm{OADP}}$, were calculated in Table II, which also included their computing times with the number of iterations, j, along with $\mathrm{TI}_{j,j-1}$ values also documented. In Table II, the results highlighted in red performed better than the original RXAD without ISS feedback, and the boldfaced results indicate the best results among all the variants including the original versions.

As shown in Table II, the best results confirmed the best result by visual inspection in Fig. 11. Except for ISSRXAD-1.1 and ISSRXAD-2.3 that had two or three AUC values lower than their counterparts of RXAD, all other five ISSRXAD performed better than the original RXAD across the board in eight detection measures.

2) CRD: Similarly, Fig. 13 shows the ADMaps of CRD along with its corresponding seven ISSCRD variants, ISSCRD-1.1, ISSCRD-1.2, ISSCRD-1.3, ISSCRD-2.1, ISSCRD-2.2, ISSCRD-2.3, and ISSCRD-3 for the HYDICE panel scene, where CRD with its seven ISSCRD variants performed comparably and detected most of the subpixel panels in the third column. To see detailed improvements in quantitative analysis, we plotted their 3-D ROC curves along with their corresponding three 2-D ROC curves in Fig. 14. Table III calculated the eight detection measures along with computing time and the required numbers of iterations, j, and $\mathrm{TI}_{j,j-1}$ values where the results highlighted by red

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

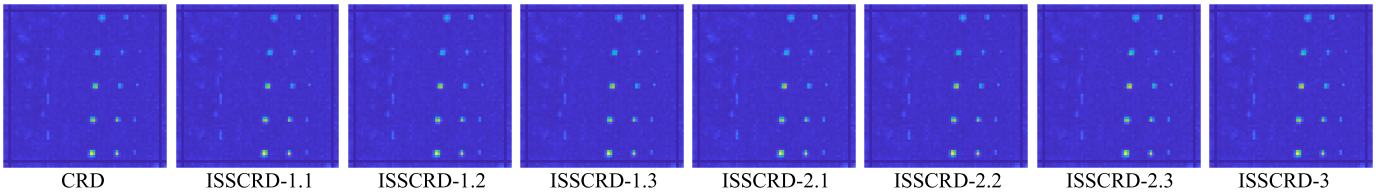


Fig. 13. ADMaps produced by CRD along with its seven ISSCRD variants on the HYDICE panel scene.

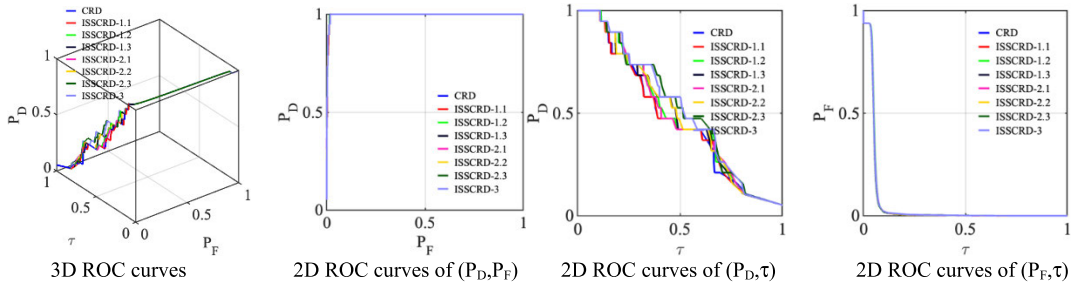


Fig. 14. 3-D ROC curves and the corresponding 2-D ROC curves produced by CRD along with its seven ISSCRD variants on the HYDICE panel scene.

TABLE III
AUC VALUES PRODUCED BY CRD AND ITS SEVEN ISSCRD VARIANTS ON THE HYDICE PANEL SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
CRD	0	0.000	0.9965	0.4450	0.9474	1.4415	1.9439	0.3924	8.4597	1.3924	3.077
ISSCRD-1.1	1	0.963	0.9965	0.4447	0.9483	1.4411	1.9448	0.3929	8.5991	1.3929	7.839
ISSCRD-1.2	2	1.000	**0.9969**	0.4681	0.9468	1.465	1.9437	0.415	8.8074	1.415	11.070
ISSCRD-1.3	1	0.963	**0.9969**	0.4671	**0.9484**	1.464	**1.9453**	0.4156	9.0562	1.4156	7.884
ISSCRD-2.1	3	0.968	0.9966	0.4676	0.9474	1.4642	1.944	0.4151	8.8976	1.4151	15.830
ISSCRD-2.2	1	0.963	0.9968	0.4716	0.9482	1.4684	1.945	0.4198	9.1083	1.4198	7.572
ISSCRD-2.3	3	0.967	0.9968	0.4977	0.9483	1.4945	1.9451	**0.4459**	**9.6241**	**1.4459**	19.506
ISSCRD-3	2	0.966	0.9968	**0.4978**	0.9476	**1.4947**	1.9444	0.4454	9.4922	1.4454	12.092

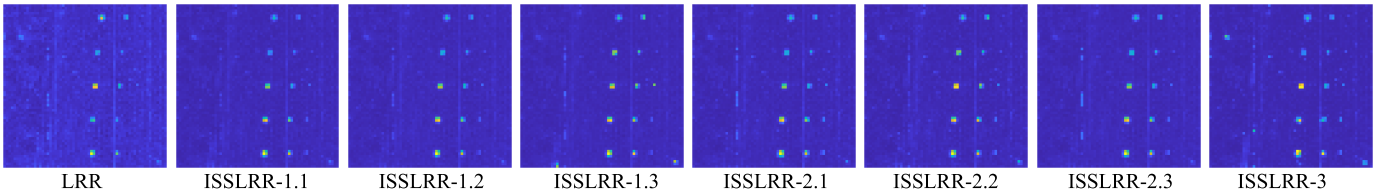


Fig. 15. ADMaps produced by LRR along with its seven ISSLRR variants on the HYDICE panel scene.

performed better than original CRD without ISS feedbacks and the boldfaced results indicate the best results among all the variants, including the original versions. According to Table III, the best results were those that included SFMaps as feedback by ISSCRD-1.3, ISSCRD-2.3, and ISSCRD-3. Since CRD already performed very well, the improvement by its ISSCRD variants was not as greater as what the ISSRXAD did for RXAD in Table II. Nevertheless, ISSCRD still managed to improve CRD to some extent across most detection measures.

3) LRR: Unlike RXAD and CRD that can be considered classic anomaly detectors, LRR is a low rank and sparse representation-based AD method that found an optimal BKG dictionary while imposing a constraint on sparse representation and then used RXAD to perform AD. Fig. 15 shows the ADMaps of LRR along with its corresponding seven ISSLRR variants, ISSLRR-1.1, ISSLRR-1.2, ISSLRR-1.3, ISSLRR-2.1, ISSLRR-2.2, ISSLRR-2.3, and ISSLRR-3 for the HYDICE panel scene, where it is clear to see that all the seven ISSCRD variants performed better than LRR with most of the subpixel panels in the third column detected. To further see quantitative analysis, we plotted their 3-D ROC curves along with their corresponding three 2-D ROC curves in Fig. 16. Table IV calculated the eight detection measures with their AUC values along with computing time and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original LRR without ISS feedbacks, and the boldfaced results indicate the best results among all the variants including the original versions. As shown in Table IV, except ISSLRR-1.2 and ISSLRR-1.1 that had one or two detection measures lower than their counterparts of LRR within a 10^{-3} error range, all other five ISSLRR variants performed better than LRR. The experiments demonstrated that LRR can be easily improved by ISSLRR.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

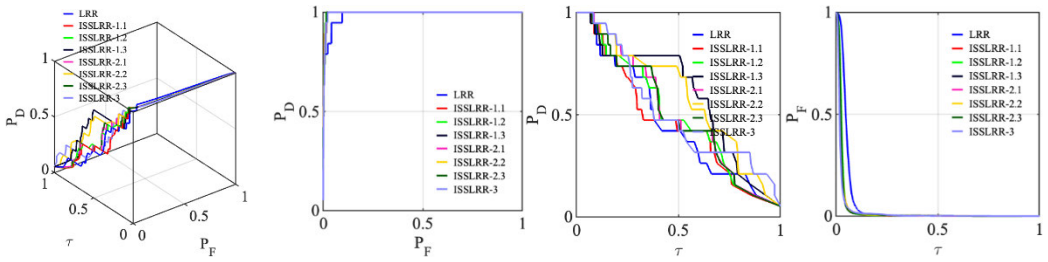


Fig. 16. 3-D ROC curves and the corresponding 2-D ROC curves produced by LRR and its seven ISSLRR variants on the HYDICE panel scene.

TABLE IV
AUC VALUES PRODUCED BY LRR AND ITS SEVEN ISSLRR VARIANTS ON THE HYDICE PANEL IMAGE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
LRR	0	0.000	0.9877	0.429	0.9481	1.4167	1.9358	0.377	8.2575	1.377	73.113
ISSLRR-1.1	2	1.000	0.9961	0.4298	0.9727	1.4259	1.9688	0.4025	15.7456	1.4025	211.823
ISSLRR-1.2	4	1.000	**0.9968**	0.4696	0.9708	1.4664	1.9675	0.4404	16.0633	1.4404	322.823
ISSLRR-1.3	13	0.969	0.996	**0.5693**	0.9731	**1.5653**	1.9691	**0.5424**	21.183	**1.5424**	1003.852
ISSLRR-2.1	7	0.970	0.9963	0.4812	0.9721	1.4775	1.9683	0.4533	17.2371	1.4533	509.481
ISSLRR-2.2	7	1.000	**0.9968**	0.5676	0.9732	1.5644	1.97	0.5408	**21.1834**	1.5408	542.333
ISSLRR-2.3	3	0.967	0.996	0.4717	0.9727	1.4676	1.9687	0.4444	17.2861	1.4444	305.859
ISSLRR-3	20	0.235	0.9954	0.4849	**0.9771**	1.4803	**1.9725**	0.4619	21.1411	1.4619	1675.929

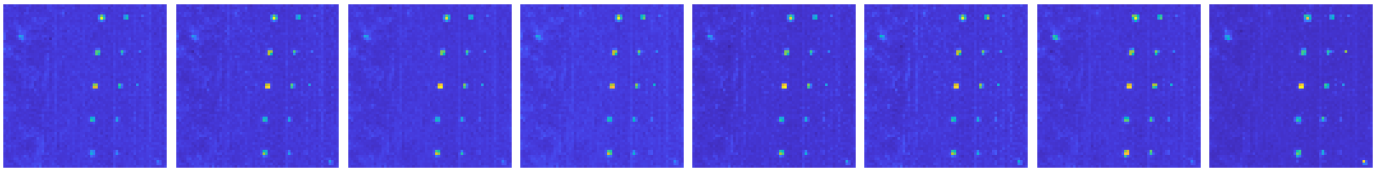


Fig. 17. ADMaps produced by LSDM-MoG along with its seven ISSLSDM-MoG variants on the HYDICE panel scene.

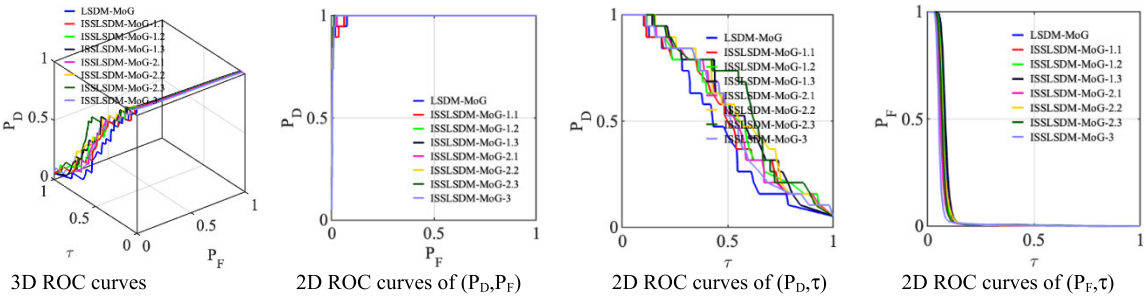


Fig. 18. 3-D ROC curves and the corresponding 2-D ROC curves produced by LSDM-MoG and its seven ISS-LSDM-MoG variants on the HYDICE Panel scene.

4) LSDM-MoG: Another AD method different from LRR is LSDM-MoG, which used a mixture of Gaussians to estimate BKG and then applied the Manhattan distance to its residual space to perform AD. Fig. 17 shows the ADMaps of LSDM-MoG along with its corresponding seven ISSLSDM-MoG variants, ISSLSDM-MoG-1.1, ISSLSDM-MoG-1.2, ISSLSDM-MoG-1.3, ISSLSDM-MoG-2.1, ISSLSDM-MoG-2.2, ISSLSDM-MoG-2.3, and ISSLSDM-MoG-3 for the HYDICE panel scene, where it is clear to see that all the seven ISSLSDM-MoG variants seemed to perform better than LSDM-MoG with most of the subpixel panels in the third column detected. To further see quantitative analysis, we plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 18. Table V calculated the eight detection measures with their AUC values along with computing time and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original LSDM-MoG without ISS feedbacks and the boldfaced results indicate the best results among all the variants, including the original versions. As shown in Table V, the clear winners were ISSLSDM-MoG-2.3 and ISSLSDM-MoG-3, both of which performed much better than LSDM-M0G. Thus, the same conclusions made on LRR were also valid for LSDM-MoG. This also indicates that LSDM-MoG can be also easily improved by ISSLSDM-MoG.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

TABLE V
AUC VALUES PRODUCED BY LSDM-MOG AND ITS SEVEN SSLSDM-MOG VARIANTS ON THE HYDICE PANEL DATA SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
LSDM-MoG	0	0.000	0.9928	0.4351	0.9213	1.4279	1.9141	0.3565	5.5304	1.3565	9.470
ISSLSDM -MoG-1.1	3	0.971	0.9929	0.4991	0.9251	1.4921	1.918	0.4242	6.6623	1.4242	44.998
ISSLSDM -MoG-1.2	4	0.972	0.9962	0.5175	0.9227	1.5137	1.9189	0.4403	6.6987	1.4403	53.652
ISSLDSM-MoG-1.3	4	1.000	0.997	0.54	0.9126	1.5369	1.9096	0.4526	6.1803	1.4526	56.545
ISSLSDM-MoG-2.1	3	0.974	0.9963	0.5132	0.9337	1.5095	1.9301	0.4469	7.7412	1.4469	45.374
ISSLSDM-MoG-2.2	3	0.974	**0.9971**	0.554	0.9233	1.5511	1.9204	0.4773	7.2218	1.4773	45.939
ISSLSDM-MoG-2.3	7	0.956	0.9965	**0.5792**	0.9266	**1.5758**	1.9231	**0.5058**	7.893	**1.5059**	111.202
ISSLSDM-MoG-3	7	0.978	0.996	0.5069	**0.9417**	1.5029	**1.9377**	0.4486	**8.6949**	1.4486	110.173

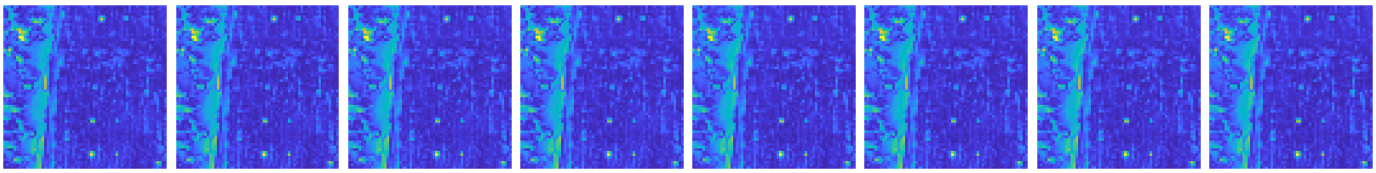


Fig. 19. ADMaps produced by PTA along with its seven ISSLRR variants on the HYDICE panel scene.

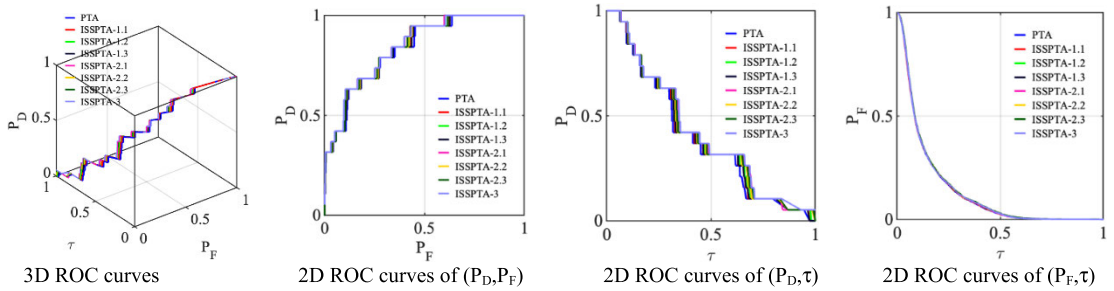


Fig. 20. 3-D ROC curves and the corresponding 2-D ROC curves produced by PTA and its seven ISSPTA variants on the HYDICE Panel scene.

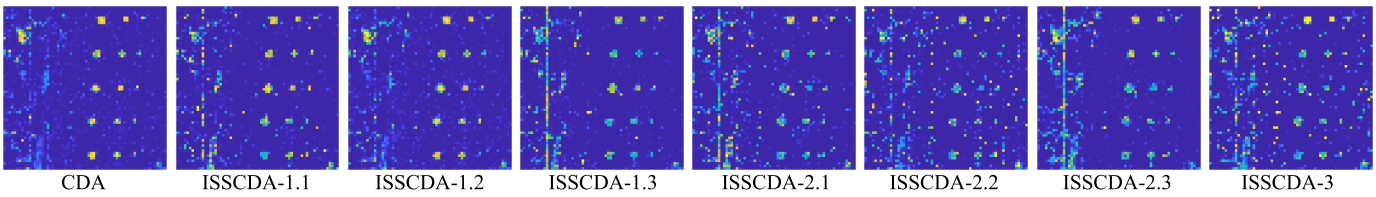


Fig. 21. ADMaps produced by RXAD using CDA along with its seven ISSCDA variants on the HYDICE panel scene.

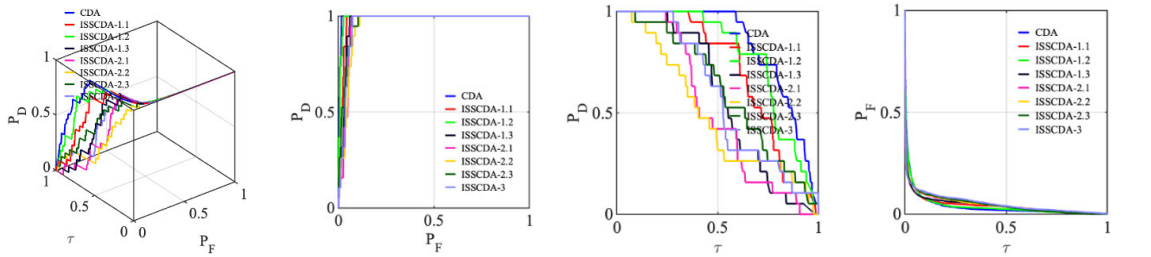
5) PTA: The fifth AD method to be tested is PTA, which used tensor approximation to decompose a data space as low-rank and sparse tensor components and then used the vector length of each data sample vector in the obtained anomaly tensor to perform AD. It has been shown to perform well on target detection but not AD [27]. Fig. 19 shows the ADMaps of PTA along with its corresponding seven ISSPTA variants, ISSPTA-1.1, ISSPTA-1.2, ISSPTA-1.3, ISSPTA-2.1, ISSPTA-2.2, ISSPTA-2.3, and ISSPTA-3 for the HYDICE panel scene, where it is clear to see that all PTAs along with its ISSPTA variants performed very poorly in detecting all panel pixels. We plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 20. Table VI calculated the eight detection measures with their AUC values along with computing time and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original PTA without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original versions. As shown in Table VI, the best one was ISSPT-2.3, which still could not detect all panel pixels well. It was the worst AD method among all the seven test AD methods. However, as will be shown in the experiments of the San Diego airport scene, this conclusion was reversed.

6) CDA: Different from all the previous five AD methods, the sixth AD method to be tested is CDA. It decomposed a data space into PC, IC, and noise subspaces and then performed AD in its IC component subspace by RXAD. Fig. 21 shows the ADMaps produced by CDA along with its corresponding seven ISSCDA variants: ISSCDA-1.1, ISSCDA-1.2, ISSCDA-1.3, ISSCDA-2.1, ISSCDA-2.2, ISSCDA-2.3, and ISSCDA-3 for the HYDICE panel scene. Unlike previous experiments, the experiments in Fig. 21 showed that all the seven ISSCDA variants seemed not to be able to improve CDA because of BKG interference introduced by ISSCDA,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
AUC VALUES PRODUCED BY PTA AND ITS SEVEN ISSPTA VARIANTS ON THE HYDICE PANEL SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
PTA	0	0.000	0.8315	0.3894	**0.8632**	1.221	1.6948	0.2527	2.8476	1.2527	48.931
ISSPTA-1.1	1	0.966	0.8366	0.3979	0.8627	1.2346	1.6993	0.2606	2.8977	1.2606	97.206
ISSPTA-1.2	1	0.962	0.8389	0.4032	0.861	1.2421	**1.6999**	0.2642	2.9013	1.2642	97.055
ISSPTA-1.3	1	0.976	0.8351	0.3984	0.861	1.2335	1.6961	0.2594	2.8661	1.2594	95.817
ISSPTA-2.1	2	0.960	0.8362	0.3812	0.8617	1.2174	1.6979	0.2429	2.7557	1.2429	150.188
ISSPTA-2.2	1	0.954	0.839	0.4072	0.8601	1.2462	1.6991	0.2674	2.9114	1.2674	99.510
ISSPTA-2.3	1	0.954	**0.8404**	**0.4111**	0.8593	**1.2516**	1.6998	**0.2705**	**2.9228**	**1.2705**	103.389
ISSPTA-3	2	0.953	0.8362	0.3863	0.8613	1.2225	1.6975	0.2476	2.7855	1.2476	156.929



3D ROC curves | 2D ROC curves of (P_D,P_F) | 2D ROC curves of (P_D,τ) | 2D ROC curves of (P_F,τ)

Fig. 22. 3-D ROC curves and the corresponding 2-D ROC curves produced by RXAD using CDA and seven ISSCDA variants on the HYDICE Panel scene.

TABLE VII
AUC VALUES PRODUCED BY RXAD USING CDA AND ITS SEVEN ISSCDA VARIANTS ON THE HYDICE PANEL SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
CDA	0	0.000	**0.9934**	**0.8268**	**0.9636**	**1.8201**	**1.957**	**0.7904**	**22.7397**	**1.7904**	0.329
ISSCDA-1.1	20	0.870	0.9801	0.7012	0.958	1.6813	1.9381	0.6592	16.6761	1.6592	24.601
ISSCDA-1.2	3	0.952	0.9907	0.7688	0.9615	1.7595	1.9522	0.7303	19.947	1.7303	5.051
ISSCDA-1.3	15	0.952	0.9707	0.5728	0.9565	1.5435	1.9271	0.5293	13.1641	1.5293	21.778
ISSCDA-2.1	20	0.881	0.9607	0.5044	0.9525	1.4651	1.9132	0.4569	10.6153	1.4569	25.471
ISSCDA-2.2	20	0.794	0.9561	0.4838	0.952	1.4399	1.9081	0.4357	10.0751	1.4357	27.619
ISSCDA-2.3	20	0.770	0.969	0.5867	0.9522	1.5557	1.9211	0.5389	12.2624	1.5389	27.863
ISSCDA-3	20	0.853	0.9639	0.5799	0.9469	1.5438	1.9109	0.5268	10.9291	1.5268	27.819

while the panel pixel detectability remains very effective. This made sense because CDA operates AD on the IC component subspace, which may still contain BKG interference, which was not removed by the PC space and was further magnified by iterative feedbacks, particularly in detection maps of ISSCDA-2.1, ISSCDA-2.2, and ISSCDA-3, where BKG is clearly affected by scattering interferers. To further see quantitative analysis, we plotted their 3-D ROC curves along with their corresponding three 2-D ROC curves in Fig. 22. Table VII calculated the eight detection measures with their AUC values along with computing time and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original CDA without ISS feedbacks, and the boldfaced results indicate the best results among all the variants including the original versions. Interestingly, compared to all the previous experiments, the values of j are very large and TI values were very low. This is because CDA performed very well, and ISSCDA made a great effort in improving CDA but could not succeed due to feedback of unwanted random interference resulting from the leakage from the PC space.

7) ICSC: The seventh AD method, referred to as ICSC, is the one developed in [27], which implemented RXAD in EAS. It should be noted that the DS was done only once at the beginning, and then, the iterative process was carried out on the sphered data without redoing DS at every iteration. Fig. 23 shows the ADMaps of RXAD using ICSC along with its corresponding seven ISSICSC variants, ISSICSC-1.1, ISSICSC-1.2, ISSICSC-1.3, ISSICSC-2.1, ISSICSC-2.2, ISSICSC-2.3, and ISSICSC-3 for the HYDICE panel scene, where all seven variants detected clearly 15 panels, including five subpixel panels in the third column by visual inspection. Specifically, ICSC worked more effectively than CDA in Fig. 21 with much better BS and much less noise. Compared to all the previous six AD methods, ICSC was clearly the best. To further conduct quantitative analysis, we plotted their 3-D ROC curves along with their corresponding three 2-D ROC

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

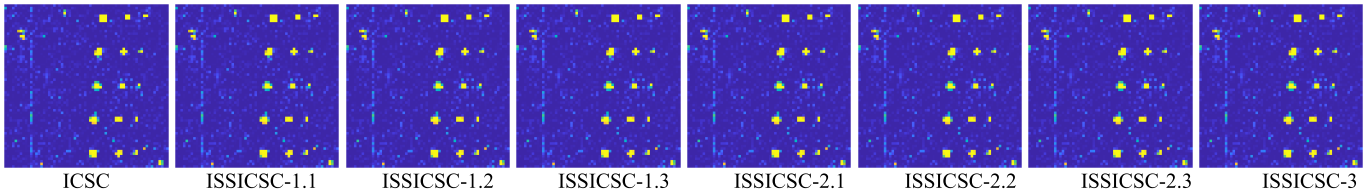


Fig. 23. ADMaps produced by RXAD using ICSC along with its seven ISSICSC variants on the HYDICE panel scene.

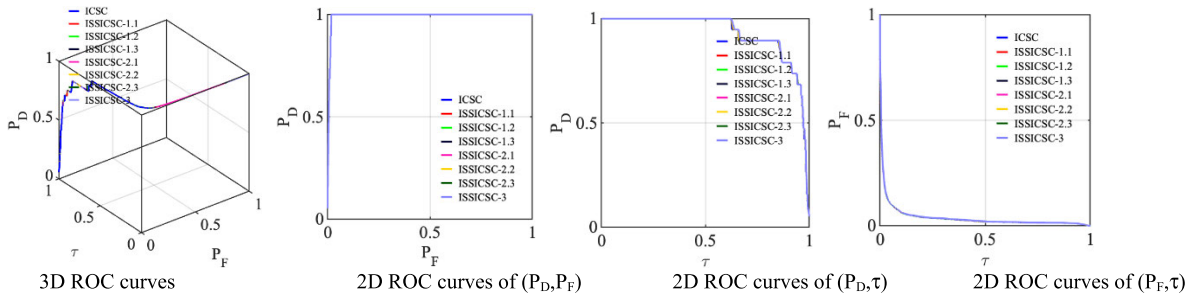


Fig. 24. 3-D ROC curves and the corresponding 2-D ROC curves produced by RXAD using ICSC and its seven ISSICSC variants on the HYDICE panel scene.

TABLE VIII
AUC VALUES PRODUCED BY RXAD USING ICSC AND ITS SEVEN ISSICSC VARIANTS ON THE HYDICE PANEL SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
RXAD	0	0.000	**0.9932**	0.9227	**0.9627**	1.9159	**1.9559**	0.8854	**24.7254**	1.8854	0.283
ISSICSC-1.1	1	0.992	**0.9932**	0.9233	0.9625	1.9165	1.9557	0.8857	24.5984	1.8857	2.434
ISSICSC-1.2	1	0.977	**0.9932**	0.9239	0.9624	1.9171	1.9557	**0.8863**	24.5986	**1.8863**	2.021
ISSICSC-1.3	1	0.992	**0.9932**	0.9231	0.9625	1.9164	1.9557	0.8856	24.6126	1.8856	2.013
ISSICSC-2.1	1	0.970	**0.9932**	0.9239	0.9622	1.9171	1.9555	0.8862	24.4664	1.8862	1.940
ISSICSC-2.2	1	0.985	**0.9932**	0.9233	0.9623	1.9165	1.9555	0.8856	24.475	1.8856	1.928
ISSICSC-2.3	1	0.970	**0.9932**	**0.924**	0.9623	**1.9172**	1.9555	**0.8863**	24.4786	**1.8863**	1.951
ISSICSC-3	1	0.963	**0.9932**	**0.924**	0.962	**1.9172**	1.9553	0.886	24.3443	1.886	1.981

curves in Fig. 24. Table VIII calculated the eight detection measures with their AUC values along with computing time and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original ICSC without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original versions. Interestingly, compared to CDA experiments, the values of j were very small, and TI values were very high. This indicated that DS was able to remove the first and second data statistics as opposed to CDA, which only removes PCs. As shown in Table VIII, ISSICSC-3 was the best.

B. HYDICE Urban Scene

Compared to the HYDICE panel scene, the HYDICE urban has less significant because anomalies are randomly distributed and have no particular spatial correlation like panels in the HYDICE panel data.

1) RXAD: Fig. 25 shows the ADMaps of RXAD along with its corresponding seven ISSRXAD variants: ISSRXAD-1.1, ISSRXAD-1.2, ISSRXAD-1.3, ISSRXAD-2.1, ISSRXAD-2.2, ISSRXAD-2.3, and ISSRXAD-3 for the HYDICE urban scene. Since there are only 21 anomalous pixels, it is difficult to evaluate their performance based on visual inspection of Fig. 25. We plotted their 3-D ROC curves along with their corresponding three 2-D ROC curves in Fig. 26. Table IX calculated eight detection measures and also documented their computing times with the number of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original RXAD without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original version. According to Table IX, all seven ISSRXAD variants did better BS than the original RXAD in AUC_{BDP}, AUC_{JBS}, and AUC_{SNPR}, and the original RXAD did not have the best value in each of the eight detection measures. On the whole, the best one was ISSRXAD-1.3, which has all the seven AUC values better than RXAD does except $AUC_{(D,F)}$, which only differs by a negligible error of 0.0003. Thus, if AD performance were based on $AUC_{(D,F)}$ values, it would have easily concluded that RXAD performed better than ISSRXAD, which is certainly misleading.

2) CRD: Fig. 27 shows the ADMaps of CRD along with its corresponding seven ISSCRD variants, ISSCRD-1.1, ISSCRD-1.2, ISSCRD-1.3, ISSCRD-2.1, ISSCRD-2.2, ISSCRD-2.3, and ISSCRD-3 for the HYDICE urban scene, where ISSCRD generally performed better than ISSCRD in Fig. 25 by visual inspection. To conduct a quantitative analysis, we plotted their 3-D ROC curves and generated their corresponding three

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

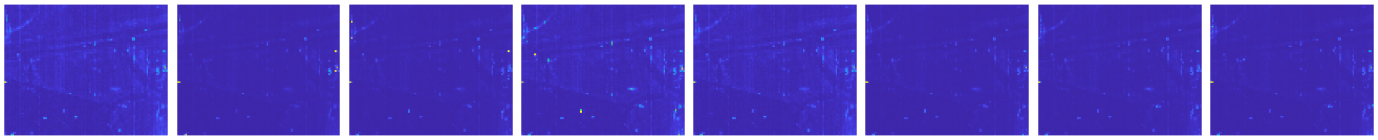


Fig. 25. ADMaps produced by RXAD along with its seven ISSRXAD variants on the HYDICE urban scene.

TABLE IX
AUC VALUES PRODUCED BY RXAD AND ITS SEVEN ISSRXAD VARIANTS ON THE HYDICE URBAN SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
RXAD	0	0.000	0.9874	0.2575	0.9641	1.2449	1.9514	0.2216	7.1647	1.2216	0.313
ISSRXAD-1.1	9	1.000	0.9873	0.144	**0.9867**	1.1313	1.9740	0.1307	**10.8207**	1.1307	27.365
ISSRXAD-1.2	10	1.000	**0.9881**	0.1254	0.9863	1.1135	**1.9744**	0.1117	9.1666	1.1117	30.879
ISSRXAD-1.3	10	0.955	0.9871	**0.2700**	0.9713	**1.2571**	1.9584	**0.2413**	9.3998	**1.2413**	31.400
ISSRXAD-2.1	1	0.992	0.9877	0.2164	0.9715	1.2041	1.9591	0.1879	7.5838	1.1879	5.443
ISSRXAD-2.2	5	1.000	0.987	0.1227	0.9865	1.1097	1.9735	0.1092	9.1203	1.1092	16.884
ISSRXAD-2.3	4	0.977	0.9862	0.13	0.9841	1.1163	1.9704	0.1142	8.1959	1.1142	14.472
ISSRXAD-3	5	1.000	0.9854	0.1292	0.9859	1.1146	1.9713	0.1151	9.1672	1.1151	17.085

TABLE X
AUC VALUES PRODUCED BY CRD AND ITS SEVEN ISSCRD VARIANTS ON THE HYDICE URBAN IMAGE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
CRD	0	0.000	**0.9504**	0.4312	0.9191	1.3816	1.8694	0.3503	5.3274	1.3503	6.066
ISSCRD-1.1	2	0.972	0.9499	0.4305	0.9225	1.3804	1.8725	0.3531	5.5579	1.3531	26.187
ISSCRD-1.2	7	1.000	0.9499	0.2904	0.9554	1.2403	1.9053	0.2458	6.5093	1.2458	68.686
ISSCRD-1.3	4	0.974	0.9486	0.4396	0.9301	1.3882	1.8787	0.3696	6.2844	1.3696	43.520
ISSCRD-2.1	6	0.966	0.9501	0.2703	**0.9604**	1.2203	**1.9105**	0.2307	6.8332	1.2307	61.179
ISSCRD-2.2	5	0.957	0.9486	**0.4531**	0.9339	**1.4017**	1.8825	**0.387**	6.8501	**1.387**	52.163
ISSCRD-2.3	6	0.964	0.947	0.2839	0.9594	1.2309	1.9063	0.2433	6.9849	1.2433	61.516
ISSCRD-3	4	0.957	0.9481	0.3408	0.9526	1.289	1.9007	0.2934	**7.1868**	1.2934	43.062

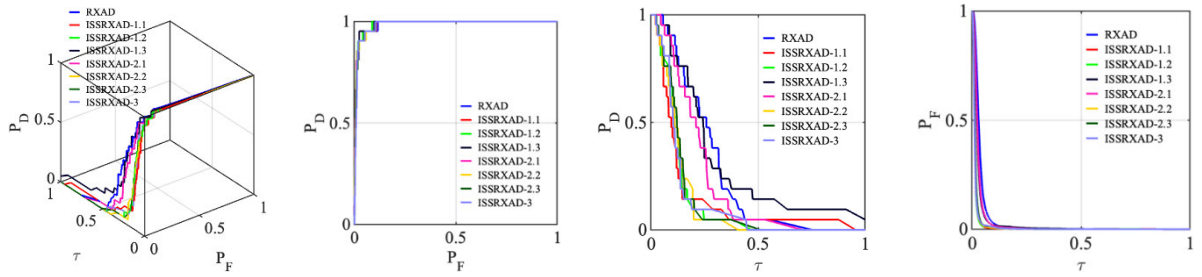


Fig. 26. 3-D ROC curves and the corresponding 2-D ROC curves produced by RXAD and its seven ISSCRD variants on the HYDICE urban scene.

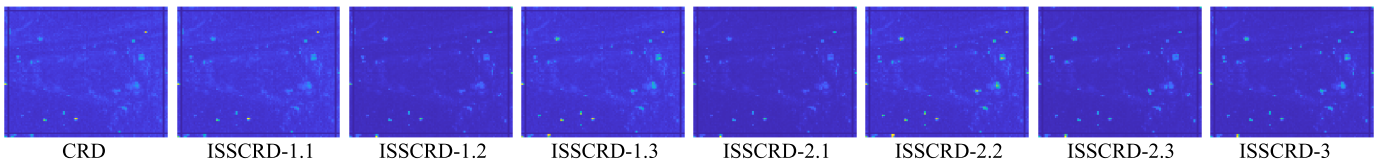
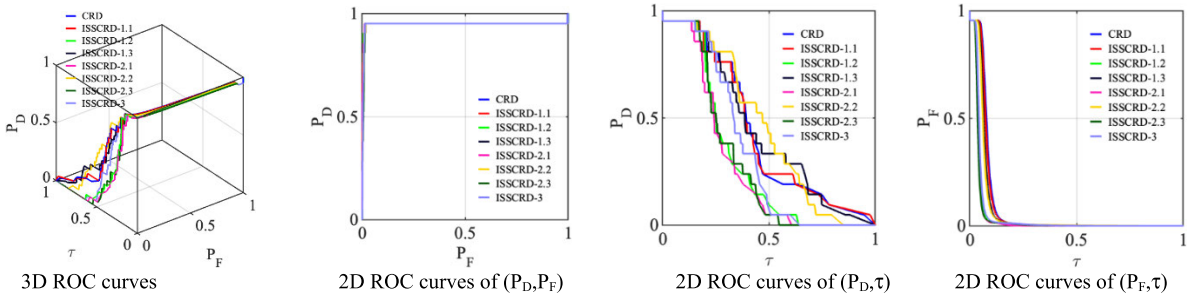


Fig. 27. ADMaps produced by CRD along with its seven ISSCRD variants on the HYDICE urban scene.

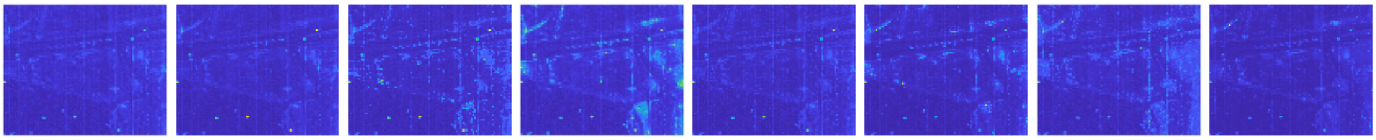
2-D ROC curves in Fig. 28. Table X calculated the eight detection measures with their AUC values and documented computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values. In Table X, the results highlighted in red performed better than the original CRD without ISS feedback, and the boldfaced results indicate the best results among all the variants, including the original version. From Table X, ISSCRD-2.2 did better AD in AUC_{ADP}, AUC_{JAD}, AUC_{ADBS}, and AUC_{OADP} than the original CRD, while the original CRD did not have the best value in each of the eight detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.



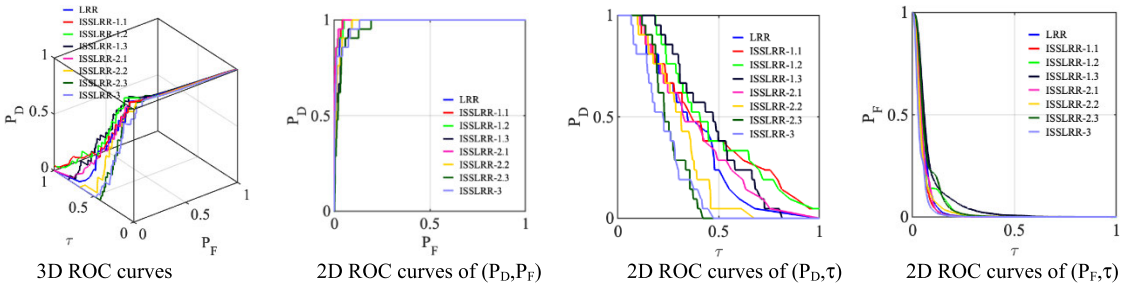
3D ROC curves | 2D ROC curves of (P_D,P_F) | 2D ROC curves of (P_D,τ) | 2D ROC curves of (P_F,τ)

Fig. 28. 3-D ROC curves and the corresponding 2-D ROC curves produced by CRD and its seven ISSCRD variants on the HYDICE urban scene.



LRR | ISSLRR-1.1 | ISSLRR-1.2 | ISSLRR-1.3 | ISSLRR-2.1 | ISSLRR-2.2 | ISSLRR-2.3 | ISSLRR-3

Fig. 29. ADMaps produced by LRR along with its seven ISSLRR variants on the HYDICE urban scene.



3D ROC curves | 2D ROC curves of (P_D,P_F) | 2D ROC curves of (P_D,τ) | 2D ROC curves of (P_F,τ)

Fig. 30. 3-D ROC curves and the corresponding 2-D ROC curves produced by LRR and its seven ISSLRR variants on the HYDICE urban scene.

TABLE XI
AUC VALUES PRODUCED BY LRR AND ITS SEVEN ISSLRR VARIANTS ON THE HYDICE URBAN SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
LRR	0	0.000	0.9924	0.3712	0.941	1.3636	1.9333	0.3122	6.288	1.3122	114.447
ISSLRR-1.1	20	0.017	0.9938	0.4245	0.9497	1.4183	1.9435	0.3742	**8.4462**	1.3742	2320.953
ISSLRR-1.2	15	0.978	0.9911	0.4539	0.9386	**1.4450**	1.9298	**0.3925**	7.3952	**1.3925**	1743.313
ISSLRR-1.3	20	0.757	0.9782	**0.4600**	0.9171	1.4382	1.8953	0.3771	5.5505	1.3771	2365.783
ISSLRR-2.1	3	0.962	**0.9949**	0.3895	0.9493	1.3844	**1.9442**	0.3388	7.6835	1.3388	452.567
ISSLRR-2.2	20	0.708	0.9816	0.3075	0.95	1.2891	1.9316	0.2575	6.145	1.2575	2303.028
ISSLRR-2.3	14	0.968	0.9677	0.2565	0.9294	1.2242	1.8972	0.1859	3.6346	1.1859	1625.617
ISSLRR-3	13	0.951	0.9810	0.2283	**0.9619**	1.2093	1.9429	0.1903	6.0009	1.1903	1560.174

measures. On the whole, the best one was ISSCRD-2.2 that has the four best AUC values compared to CRD, which has only one best AUC value, $AUC_{(D,F)}$. Once again, if only $AUC_{(D,F)}$ value were used to evaluate AD performance, we would have mistakenly concluded that CRD performed better than ISSCRD, which is certainly not true.

3) LRR: Fig. 29 shows the ADMaps of LRR along with its corresponding seven ISSLRR variants, ISSLRR-1.1, ISSLRR-1.2, ISSLRR-1.3, ISSLRR-2.1, ISSLRR-2.2, ISSLRR-2.3, and ISSLRR-3 for the HYDICE urban scene, where ISSLRR generally performed better than ISSRXAD in Fig. 25 by visual inspection. To conduct quantitative analysis, we plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 30. Table XI calculated the eight detection measures and their AUC values along with computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original LRR without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original version. Like ISSCRD-2.2 in Table X, ISSLRR-1.2 in Table XI was the best one in the sense that it has the three best AUC values in AD, AUC_{JAD}, AUC_{ADBS}, and AUC_{OADP}, compared to LRR, which did not have best values in each of eight detection measures. Nevertheless, LRR has also the second highest $AUC_{(D,F)}$ value. Consequently, a misleading conclusion is also very likely made for LRR, which performed better than all the other six ISSLRR variants.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

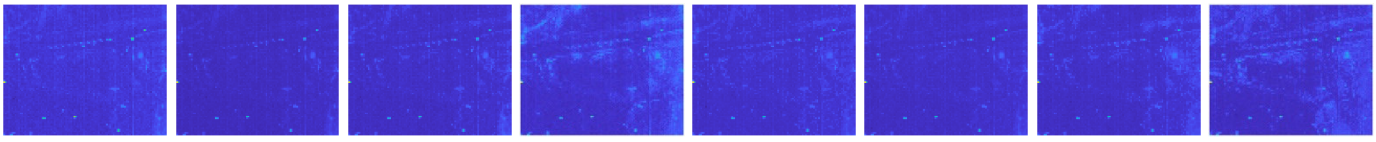


Fig. 31. ADMaps produced by LSDM-MoG along with its seven ISSLSDM-MoG variants on the HYDICE urban scene.

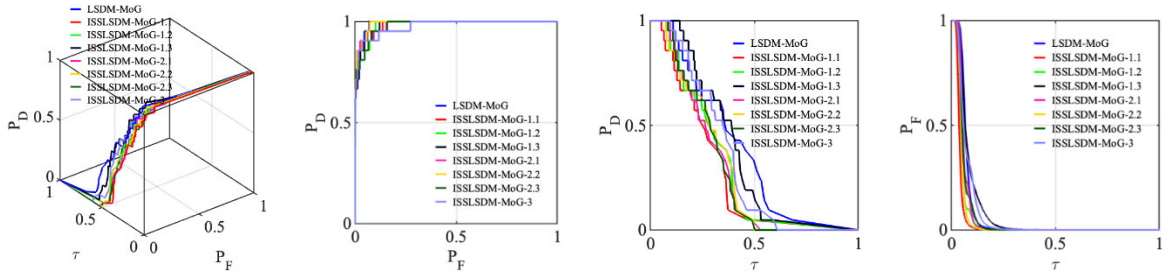


Fig. 32. 3-D ROC curves and the corresponding 2-D ROC curves by LSDM-MoG and seven ISSLSDM-MoG variants on the HYDICE urban scene.

TABLE XII
AUC VALUES PRODUCED BY LSDM-MOG AND ITS SEVEN ISSLSDM-MOG VARIANTS ON THE HYDICE URBAN SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
LSDM-MoG	0	0.000	**0.9915**	**0.3741**	0.9281	**1.3656**	1.9196	**0.3022**	5.2053	**1.3022**	15.039
ISSLSDM -MoG-1.1	13	0.970	0.9873	0.2596	**0.9598**	1.2469	**1.9471**	0.2194	**6.4622**	1.2194	276.248
ISSLSDM -MoG-1.2	4	0.956	0.9862	0.288	0.948	1.2742	1.9342	0.2359	5.5336	1.2359	110.401
ISSLDSM-MoG-1.3	20	0.856	0.984	0.3636	0.9175	1.3476	1.9015	0.281	4.4058	1.281	497.854
ISSLSDM-MoG-2.1	14	0.959	0.9849	0.2683	0.9389	1.2532	1.9238	0.2072	4.3935	1.2072	349.986
ISSLSDM-MoG-2.2	3	1.000	0.991	0.2787	0.9498	1.2697	1.9408	0.2285	5.5561	1.2285	83.931
ISSLSDM-MoG-2.3	8	0.967	0.9814	0.2738	0.9407	1.2551	1.922	0.2144	4.6155	1.2144	224.694
ISSLSDM-MoG-3	17	0.965	0.977	0.3249	0.9273	1.3019	1.9043	0.2522	4.4707	1.2522	455.252

4) LSDM-MoG: Fig. 31 shows the ADMaps of LSDM-MoG along with its corresponding seven ISSLSDM-MoG variants, ISSLSDM-MoG-1.1, ISSLSDM-MoG-1.2, ISSLSDM-MoG-1.3, ISSLSDM-MoG-2.1, ISSLSDM-MoG-2.2, ISSLSDM-MoG-2.3, and ISSLSDM-MoG-3 for the HYDICE urban scene, where ISSLSDM-MoG generally performed better than ISSRXAD in Fig. 25 by visual inspection. To conduct quantitative analysis, we plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 32. Table XII calculated the eight detection measures with their AUC values and also documented the computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original LSDM-MoG without ISS feedbacks, and the boldfaced results indicate the best results among all the variants including the original version. There were two interesting findings from Table XII. One is that LSDM-MoG without ISS feedback performed the best in AD in the sense of producing the highest values of AUC_{ADP}, AUC_{JAD}, AUC_{ADBS}, and AUC_{OADP}. This is mainly due to its effective removal of Gaussian BKG by MoG. A second one is that ISSLSDM-MoG-1.1 in Table XII was the best one in BS to produce the highest values of AUC_{BDP}, AUC_{JBS}, and AUC_{SNPR}. That is, despite that LSDM-MoG did the best in AD by removing Gaussian BKG, it could not effectively suppress BKG. This indicated that BKG in the scene was more complicated than the Gaussian BKG.

5) PTA: Fig. 33 shows the ADMaps of PTA along with its corresponding seven ISSPTA variants, ISSPTA-1.1, ISSPTA-1.2, ISSPTA-1.3, ISSPTA-2.1, ISSPTA-2.2, ISSPTA-2.3, and ISSPTA-3 for the HYDICE urban scene, where ISSPTA performed very poorly by visual inspection and much worse than RXAD in Fig. 25, CRD in Fig. 27, LRR in Fig. 29, and LSDM-MoG in Fig. 31. To conduct quantitative analysis, we plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 34. Table XIII calculated the eight detection measures with their AUC values and also documented the computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original PTA without ISS feedbacks, and the boldfaced results indicate the best results among all the variants including the original version. According to Table XIII, three interesting findings are noteworthy. It was very clear that ISSPTA-3 was the best across aboard in all eight detection measures. Another is that all the seven ISSPTA variants performed much better than PTA in all

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

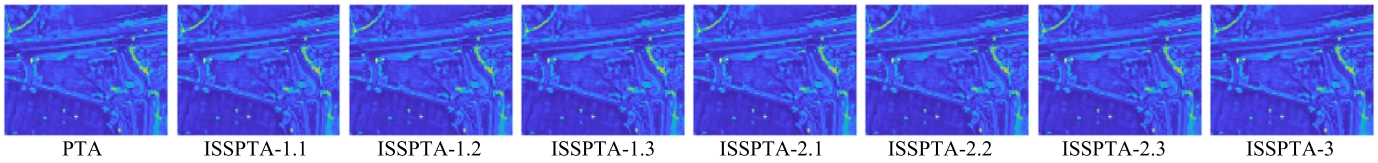


Fig. 33. ADMaps produced by PTA along with its seven ISSPTA variants on the HYDICE urban scene.

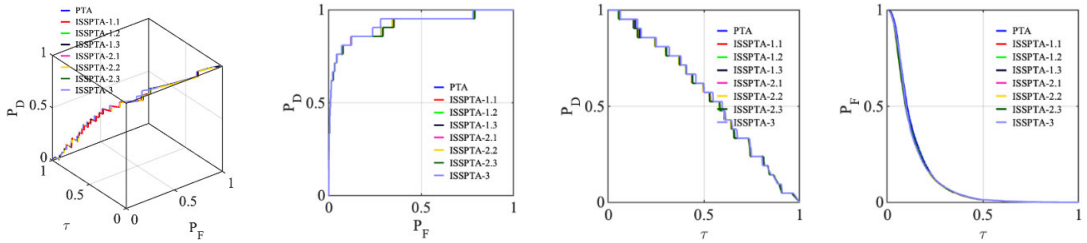


Fig. 34. 3-D ROC curves and the corresponding 2-D ROC curves produced by PTA and its seven ISSPTA variants on the HYDICE urban image.

TABLE XIII
AUC VALUES PRODUCED BY PTA AND ITS SEVEN ISSPTA VARIANTS ON THE HYDICE URBAN SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
PTA	0	0.000	0.9160	0.5504	0.8617	1.4664	1.7777	0.4121	3.9807	1.4121	79.184
ISSPTA-1.1	1	0.973	0.9156	0.5469	0.8683	1.4625	1.7839	0.4152	4.1528	1.4152	162.439
ISSPTA-1.2	1	0.982	0.9150	0.5482	0.8669	1.4632	1.7819	0.4151	4.1187	1.4151	162.088
ISSPTA-1.3	1	0.984	0.9156	0.5475	0.8681	1.4631	1.7837	0.4156	4.1498	1.4156	165.211
ISSPTA-2.1	1	0.975	0.9149	0.5476	0.8687	1.4625	1.7836	0.4163	4.171	1.4163	164.026
ISSPTA-2.2	1	0.980	0.9155	0.5470	**0.8692**	1.4624	1.7846	0.4161	**4.1806**	1.4161	163.217
ISSPTA-2.3	1	0.970	0.9149	0.5480	0.8684	1.4630	1.7833	0.4164	4.1633	1.4164	163.930
ISSPTA-3	2	0.977	**0.9219**	**0.5556**	0.8663	**1.4775**	**1.7881**	**0.4218**	4.1541	**1.4218**	245.433

TABLE XIV
AUC VALUES PRODUCED BY RXAD USING CDA AND SEVEN ISSCDA VARIANTS ON THE HYDICE URBAN SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
CDA	0	0.000	0.9564	0.3791	**0.9793**	1.3356	1.9357	0.3584	18.2933	1.3584	0.526
ISSCDA-1.1	20	0.894	0.9675	0.3873	0.9767	1.3548	1.9442	0.364	16.6216	1.364	66.811
ISSCDA-1.2	20	0.895	0.9667	0.3663	0.9767	1.333	1.9434	0.3429	15.6938	1.3429	66.440
ISSCDA-1.3	18	0.959	0.9600	0.4984	0.9770	1.4584	1.9371	0.4755	21.7081	1.4755	59.941
ISSCDA-2.1	20	0.922	**0.9819**	0.6923	0.9751	1.6741	**1.9569**	0.6673	27.7671	1.6673	68.207
ISSCDA-2.2	20	0.687	0.9539	0.447	0.9762	1.4009	1.9301	0.4232	18.7648	1.4232	67.673
ISSCDA-2.3	20	0.803	0.9700	0.5196	0.9751	1.4896	1.9451	0.4947	20.8832	1.4947	68.952
ISSCDA-3	20	0.782	**0.9819**	**0.7235**	0.9744	**1.7054**	1.9563	**0.6979**	**28.2613**	**1.6979**	67.224

other measures that were the best except $AUC_{(D,F)}$, AUC_{ADP}, and AUC_{JAD}. A third one is that PTA was the second best in producing $AUC_{(D,F)}$, AUC_{ADP}, and AUC_{JAD}. In other words, if we only base on anomaly detectability, PTA performed very well. The truth of the matter is that PTA was, in fact, the worst one in BKG suppressibility. These experiments demonstrated that, in evaluating AD performance, we need to look into both anomaly detectability and BKG suppressibility.

6) CDA: Fig. 35 shows the ADMaps produced by CDA using RXAD along with its corresponding seven ISSCDA variants, ISSCDA-1.1, ISSCDA-1.2, ISSCDA-1.3, ISSCDA-2.1, ISSCDA-2.2, ISSCDA-2.3, and ISSCDA-3 for the HYDICE urban scene, where it is difficult to evaluate their performances by visual inspection. In this case, we plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 36. Table XIV calculated the eight detection measures with their AUC values and also documented the computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original CDA without ISS feedbacks, and the boldfaced results indicate the best results among all the variants including the original version. Based on the results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

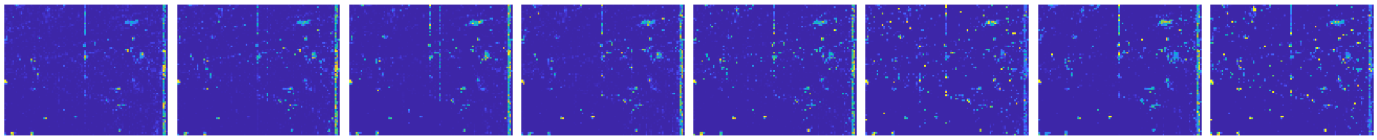


Fig. 35. ADMaps produced by RXAD using CDA along with its seven ISSCDA variants on the HYDICE urban scene.

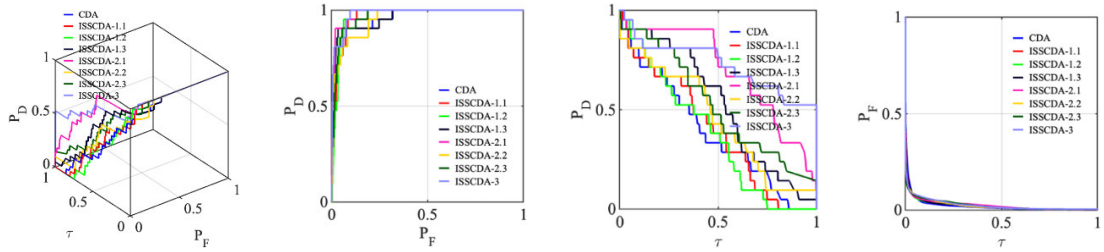


Fig. 36. 3-D ROC curves and the corresponding 2-D ROC curves produced by RXAD using CDA and seven ISSCDA variants on the HYDICE urban scene.

TABLE XV
AUC VALUES PRODUCED BY RXAD USING ICSC AND ITS SEVEN ISSICSC VARIANTS ON THE HYDICE URBAN SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
ICSC	0	0.000	0.9633	0.5188	0.9789	1.4822	1.9423	0.4978	24.6307	1.4978	0.357
ISSICSC-1.1	1	0.980	0.9634	0.5222	0.9788	1.4856	1.9422	0.501	24.658	1.501	5.657
ISSICSC-1.2	1	0.995	0.9634	0.5231	0.9788	1.4865	1.9422	0.5019	24.6976	1.5019	5.650
ISSICSC-1.3	1	0.976	0.9634	0.5213	**0.9791**	1.4847	**1.9425**	0.5004	**24.8955**	1.5004	5.852
ISSICSC-2.1	2	0.968	0.9633	0.5249	0.9785	1.4882	1.9417	0.5034	24.3758	1.5034	8.232
ISSICSC-2.2	1	0.980	0.9632	0.5228	0.9789	1.486	1.9421	0.5017	24.8277	1.5017	5.632
ISSICSC-2.3	1	0.957	**0.9635**	0.5239	0.9789	1.4874	1.9424	0.5028	24.8762	1.5028	5.840
ISSICSC-3	2	0.973	**0.9635**	**0.527**	0.9785	**1.4905**	1.942	**0.5054**	24.4597	**1.5054**	9.070

in Table XIV, CDA had a problem with BKG interference caused by feedback since the number of iterations was very high, and TI values were low. Nevertheless, CDA was the best to produce the highest value of AUC_{BDP} due to the fact that CDA used PCs to remove BKG, including Gaussian BKG. This explained why CDA could not remove BKG effectively by simply using PCs. However, except for BDP and a few detection measures, all seven ISSCDAs performed better than CDA. It also clearly showed that ISSCDA-3 was the best one in all seven detection measures except BDP.

7) ICSC: Fig. 37 shows the ADMaps of RXAD using ICSC along with its corresponding seven ISSICSC variants, ISSICSC-1.1, ISSICSC-1.2, ISSICSC-1.3, ISSICSC-2.1, ISSICSC-2.2, ISSICSC-2.3, and ISSICSC-3 for the HYDICE urban scene, where it was difficult to conclude which one was best. To further see their differences, we plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 38. Table XV calculated the eight detection measures with their AUC values and also documented the computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original ICSC without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original version. Interestingly, compared to CDA in Table XIV, the values of j were very small, and TI values in Table XV were very high. As shown in Table XV, ISSICSC-2.1 was the best since it produced the best four AUC values out of eight detection measures. If we further compared Tables XIV and XV, it was surprising to see that the conclusions made for CDA and ICSC for the HYDICE urban scene were reversed. That is, CDA produced better results in Table XIV than ICSC did in Table XV. Nevertheless, ISSICSC seems better than ISSCDA in the sense of its stability caused by feedback.

C. AVIRIS San Diego Airport Scene

The San Diego airport scene is quite different form previous HYDICE panels and urban scenes in terms of anomalies to be detected where the anomalies considered as the three airplanes are composed of tens of pixels and very visible in their large sizes compared to the latter which contains anomalies made up of a few pixels and are barely visible. According to [27], the San Diego airport scene is not suitable for AD. Consequently, the conclusions drawn from this scene are also expected to be different from that made from previous HYDICE experiments.

1) RXAD: Fig. 39 shows the ADMaps of RXAD along with its corresponding seven ISSRXAD variants, ISSRXAD-1.1, ISSRXAD-1.2, ISSRXAD-1.3, ISSRXAD-2.1, ISSRXAD-2.2, ISSRXAD-2.3, and ISSRXAD-3 for the San Diego airport scene, where no airplanes could be found in all the detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

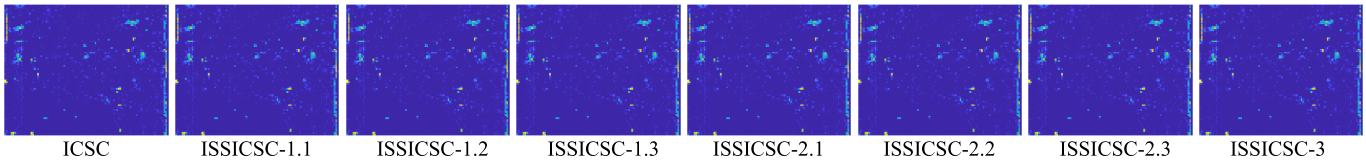


Fig. 37. ADMaps produced by RXAD using ICSC along with its seven ISSICSC variants on the HYDICE urban scene.

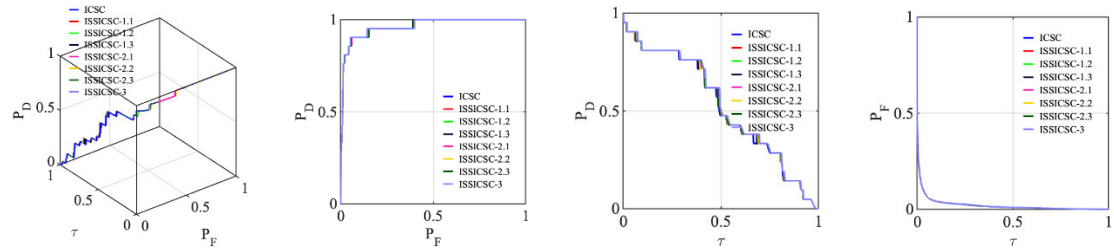


Fig. 38. 3-D ROC curves and the corresponding 2-D ROC curves produced by RXAD using ICSC and its seven ISSICSC variants on the HYDICE urban scene.

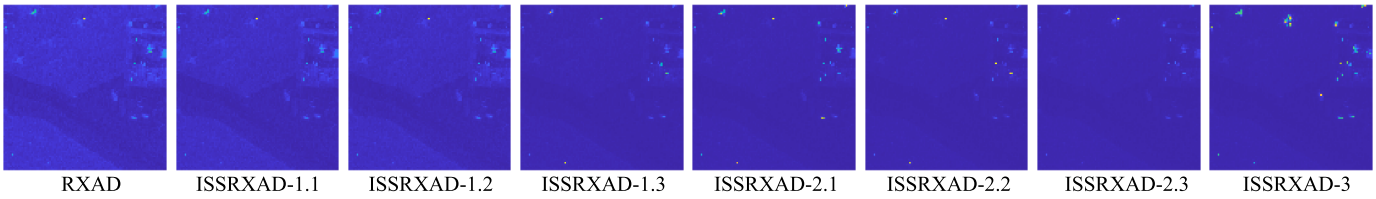


Fig. 39. ADMaps produced by RXAD along with its seven ISS-AD variants on the San Diego airport scene.

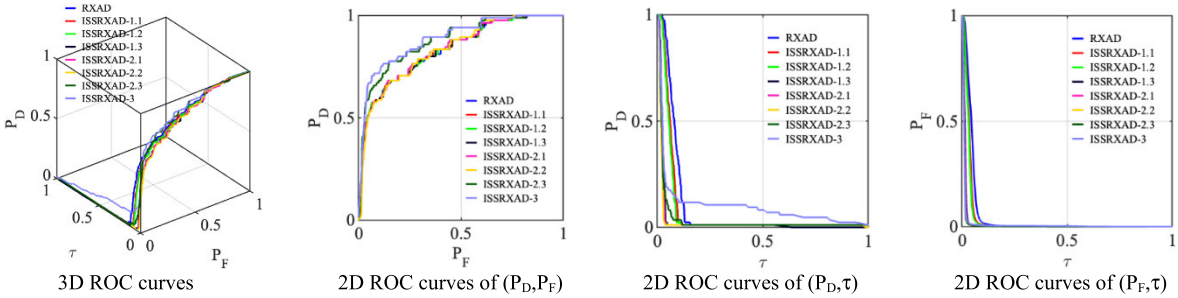


Fig. 40. 3-D ROC curves and the corresponding 2-D ROC curves produced by RXAD and its seven ISSRXAD variants on the AVIRIS San Diego airport scene.

TABLE XVI
AUC VALUES PRODUCED BY RXAD AND SEVEN ISSRXAD VARIANTS ON THE AVIRIS SAN DIEGO AIRPORT SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
RXAD	0	0.000	0.8314	0.0802	0.9547	0.9116	1.7861	0.0349	1.7711	1.0349	0.372
ISSRXAD-1.1	1	0.982	0.8325	0.0607	0.9656	0.8933	1.7982	0.0264	1.7676	1.0264	7.290
ISSRXAD-1.2	1	1.000	0.8314	0.0554	0.9686	0.8868	1.8	0.0240	1.7665	1.024	7.460
ISSRXAD-1.3	6	1.000	0.833	0.0307	0.9859	0.8637	1.819	0.0166	2.1813	1.0166	26.054
ISSRXAD-2.1	8	1.000	0.8342	0.0209	0.9871	0.8551	1.8213	0.0079	1.6125	1.0079	33.954
ISSRXAD-2.2	6	1.000	0.8361	0.0295	0.9893	0.8656	1.8254	0.0187	2.7501	1.0187	26.379
ISSRXAD-2.3	3	1.000	0.8780	0.0255	**0.9895**	0.9035	1.8675	0.0150	2.4346	1.015	14.988
ISSRXAD-3	13	0.964	**0.8869**	**0.0883**	0.9872	**0.9751**	**1.8741**	**0.0755**	**6.9168**	**1.0755**	51.821

maps. This indicated that airplanes were not considered anomalies by RXAD. We then plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 40. Table XVI calculated the eight detection measures with their AUC values and also documented their computing times and the required numbers of iterations, j, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

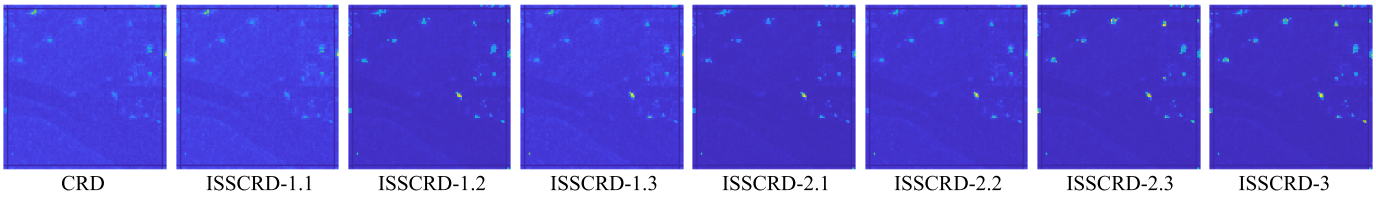


Fig. 41. ADMaps produced by CRD along with its seven ISS-AD variants on the San Diego airport scene.

TABLE XVII
AUC VALUES PRODUCED BY CRD AND ITS SEVEN ISSCRD VARIANTS ON THE AVIRIS SAN DIEGO AIRPORT SCENE

Anomaly Detector	j	$\mathrm{TI}_{j,j-1}$	$\mathrm{AUC}_{(D,F)}$	$\mathrm{AUC}_{\mathrm{ADP}}$	$\mathrm{AUC}_{\mathrm{BDP}}$	$\mathrm{AUC}_{\mathrm{JAD}}$	$\mathrm{AUC}_{\mathrm{JBS}}$	$\mathrm{AUC}_{\mathrm{ADBS}}$	$\mathrm{AUC}_{\mathrm{SNPR}}$	$\mathrm{AUC}_{\mathrm{OADP}}$	Time(s)
CRD	0	0.000	0.9063	0.1609	0.9076	1.0672	1.8139	0.0685	1.7407	1.0685	7.747
ISSCRD-1.1	2	0.985	0.9157	0.1659	0.9082	1.0815	1.8239	0.0741	1.8066	1.0741	34.108
ISSCRD-1.2	6	0.976	0.9048	0.1441	0.952	1.0488	1.8567	0.096	2.9985	1.096	78.745
ISSCRD-1.3	5	0.957	0.9196	0.143	0.9278	1.0626	1.8474	0.0708	1.9801	1.0708	68.412
ISSCRD-2.1	6	1.000	0.9151	0.1088	**0.9609**	1.0238	1.876	0.0697	2.7847	1.0697	81.735
ISSCRD-2.2	4	0.951	0.9213	0.1342	0.9449	1.0555	1.8663	0.0791	2.4366	1.0791	57.760
ISSCRD-2.3	6	0.970	0.9267	**0.173**	0.9578	**1.0996**	1.8845	**0.1308**	4.0991	**1.1308**	81.439
ISSCRD-3	7	0.952	**0.9287**	0.1659	0.9603	1.0946	**1.889**	0.1262	**4.176**	1.1262	92.520

$\mathrm{TI}_{j,j-1}$ values, where the results highlighted by red performed better than original RXAD without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original version. Table XVI further confirmed the fact that airplanes were indeed not anomalies since their detection rates, $\mathrm{AUC}_{\mathrm{ADP}}$, and $\mathrm{AUC}_{\mathrm{ADBS}}$ were very low and less than 0.09. On the contrary, the BS values were very high. Specifically, all seven ISSRXAD variants did better BS than the original RXAD in $\mathrm{AUC}_{\mathrm{BDP}}$ and $\mathrm{AUC}_{\mathrm{JBS}}$. Also, RXAD did not have the best value in each of the eight detection measures. On the whole, the best one was ISSRXAD-3, which had all seven AUC values better than RXAD.

2) CRD: Fig. 41 shows the ADMaps of CRD along with its corresponding seven ISSCRD variants, ISSCRD-1.1, ISSCRD-1.2, ISSCRD-1.3, ISSCRD-2.1, ISSCRD-2.2, ISSCRD-2.3, and ISSCRD-3 for the San Diego airport scene, where CRD and its ISSCRD performed much better than CRD in the sense that three detected airplanes were barely visible in Fig. 39. We then plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 42. Table XVII calculated the eight detection measures with their AUC values and also documented the computing times and the required numbers of iterations, j, and $\mathrm{TI}_{j,j-1}$ values, where the results highlighted by red performed better than original CRD without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original version. Although CRD and its ISSCRD variants improved RXAD in the detection of airplanes, Table XVII showed that the detection rates of three airplanes, $\mathrm{AUC}_{\mathrm{ADP}}$, and $\mathrm{AUC}_{\mathrm{ADBS}}$ by ISSCRD were still very low and less than 0.20. Specifically, except $\mathrm{AUC}_{\mathrm{ADP}}$ and $\mathrm{AUC}_{\mathrm{JAD}}$, all seven ISSCRD variants did better than the original CRD in all eight detection measures with the exception of $\mathrm{AUC}_{(D,F)}$ by ISSCRD-1.2. Also, the original CRD did not have the best value in each of the eight detection measures. Overall speaking, the best ones were ISSCRD-2.3 and ISSCRD-3, which had four or three highest AUC values out of eight detection measures.

3) LRR: Fig. 43 shows the ADMaps of LRR along with its corresponding seven ISSLRR variants, ISSLRR-1.1, ISSLRR-1.2, ISSLRR-1.3, ISSLRR-2.1, ISSLRR-2.2, ISSLRR-2.3, and ISSLRR-3 for the San Diego airport scene, where LRR and its ISSLRR also performed much better than RXAD but comparably to CRD in Fig. 41 by visual inspection. We then plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 44. Table XVIII calculated the eight detection measures with their AUC values and also documented the computing times and the required numbers of iterations, j, and $\mathrm{TI}_{j,j-1}$ values, where the results highlighted by red performed better than original LRR without ISS feedbacks and, the boldfaced results indicate the best results among all the variants, including the original version. Although LRR and its ISSLRR variants improved the detection of airplanes, Table XVIII showed that, except ISSLRR-2.3, $\mathrm{AUC}_{\mathrm{ADP}}$ and $\mathrm{AUC}_{\mathrm{ADBS}}$ of the three airplanes were still very low and less than 0.1. Specifically, none of all eight detection measures produced by LRR was the best value. Overall speaking, ISSLRR performed slightly worse than ISSCRD, and the best one was ISSLRR-2.3, which had the five highest AUC values out of eight detection measures.

4) LSDM-MoG: Fig. 45 shows the ADMaps of LSDM-MoG along with its corresponding seven ISSLSDM-MoG variants, ISSLSDM-MoG-1.1, ISSLSDM-MoG-1.2, ISSLSDM-MOG-1.3, ISSLSDM-MoG-2.1, ISSLSDM-MoG-2.2, ISSLSDM-MoG-2.3, and ISSLSDM-MoG-3 for the San Diego airport scene, where LSDM-MoG and its ISSLSDM-MoG also performed much better than RXAD in Fig. 39, CRD in Fig. 41, and LRR in Fig. 43 by visual inspection

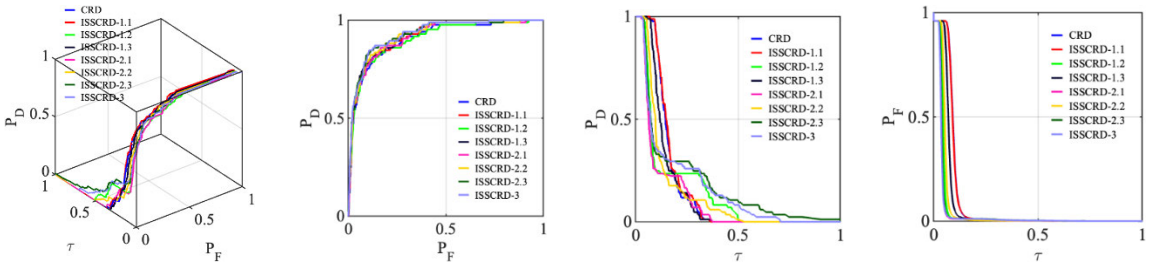
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

TABLE XVIII
AUC VALUES PRODUCED BY LRR AND ITS SEVEN ISSLRR VARIANTS ON THE AVIRIS SAN DIEGO AIRPORT SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
LRR	0	0.000	0.8789	0.1251	0.9314	1.004	1.8103	0.0565	1.8232	1.0565	129.178
ISSLRR-1.1	10	0.970	0.8644	0.0485	0.9723	0.9129	1.8367	0.0208	1.7486	1.0208	1468.820
ISSLRR-1.2	8	0.957	0.8112	0.0667	0.9658	0.8779	1.7771	0.0325	1.952	1.0325	1136.935
ISSLRR-1.3	20	0.816	0.9205	0.1003	0.9665	1.0207	1.887	0.0668	2.9971	1.0668	2769.380
ISSLRR-2.1	16	0.964	0.8895	0.0718	**0.9737**	0.9613	1.8632	0.0455	2.7316	1.0455	2133.924
ISSLRR-2.2	20	0.831	**0.9232**	0.1000	0.9704	1.0232	**1.8936**	0.0704	3.3825	1.0704	2700.186
ISSLRR-2.3	16	0.954	0.9082	**0.1687**	0.9531	**1.077**	1.8613	**0.1218**	**3.5943**	**1.1218**	2215.359
ISSLRR-3	13	0.956	0.918	0.1054	0.968	1.0233	1.8859	0.0733	3.2885	1.0733	1840.604

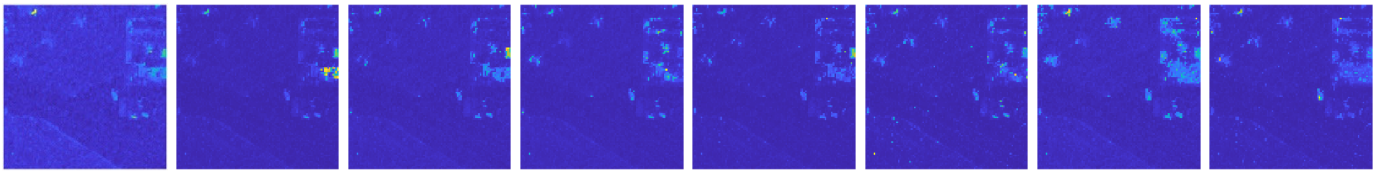
TABLE XIX
AUC VALUES PRODUCED BY LSDM-MOG AND ITS SEVEN ISSLSDM-MOG VARIANTS ON THE AVIRIS SAN DIEGO AIRPORT SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
LSDM-MoG	0	0.000	0.8403	0.2111	0.8571	1.0513	1.6973	0.0681	1.4765	1.0681	9.730
ISSLSDM -MoG-1.1	16	0.953	0.9034	0.2025	0.8937	1.106	1.7971	0.0962	1.9051	1.0962	387.793
ISSLSDM -MoG-1.2	4	0.985	0.8375	0.165	0.9058	1.0025	1.7433	0.0708	1.751	1.0708	101.452
ISSLDSM-MoG-1.3	14	0.953	0.8792	0.2415	0.9044	1.1208	1.7836	0.1459	2.5255	1.1459	365.233
ISSLSDM-MoG-2.1	9	0.994	0.9052	0.24	0.9143	1.1452	1.8195	0.1543	2.8008	1.1543	269.450
ISSLSDM-MoG-2.2	13	0.960	**0.9359**	0.244	0.9083	1.1799	1.8442	0.1523	2.66	1.1523	357.623
ISSLSDM-MoG-2.3	11	0.957	0.9183	**0.264**	0.9093	**1.1823**	1.8276	**0.1732**	**2.9089**	**1.1732**	329.172
ISSLSDM-MoG-3	6	0.956	0.9202	0.1877	**0.9345**	1.1079	**1.8547**	0.1222	2.8664	1.1222	187.998



3D ROC curves | 2D ROC curves of (P_D,P_F) | 2D ROC curves of (P_D,τ) | 2D ROC curves of (P_F,τ)

Fig. 42. 3-D ROC curves and the corresponding 2-D ROC curves produced by CRD and its seven ISSCRD variants on the AVIRIS San Diego airport scene.

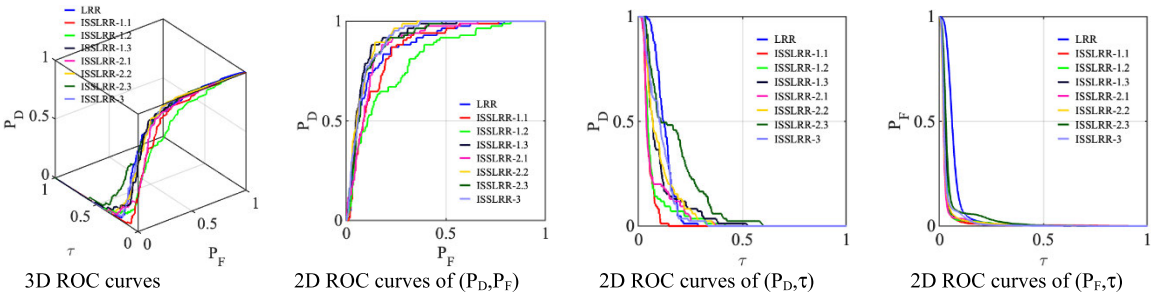


LRR | ISSLRR-1.1 | ISSLRR-1.2 | ISSLRR-1.3 | ISSLRR-2.1 | ISSLRR-2.2 | ISSLRR-2.3 | ISSLRR-3

Fig. 43. ADMaps produced by LRR along with its seven ISS-AD variants on the San Diego airport scene.

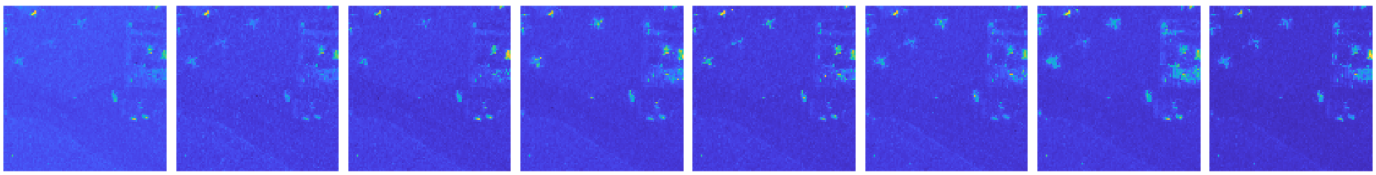
in the sense that three detected airplanes were more visible in Fig. 45. We then plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 46. Table XIX calculated the eight detection measures with their AUC values and also documented the computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original LSDM-MoG without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original version. Although LSDM-MoG and its ISSLSDM-MoG variants also improved the detection of airplanes over ISSLRR and ISSRXAD to some extent, AUC_{ADP} and AUC_{ADBS} of three airplanes were still low and little more than 0.2. Interestingly, ISSLSDM-MoG did better in AUC_{ADBS} than ISSLRR but worse than ISSLRR in AUC_{BDP}. Once again, none of all eight detection measures produced by LSDM-MoG was the best value. On the whole, the best ones were ISSLSDM-MoG-2.3 and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.



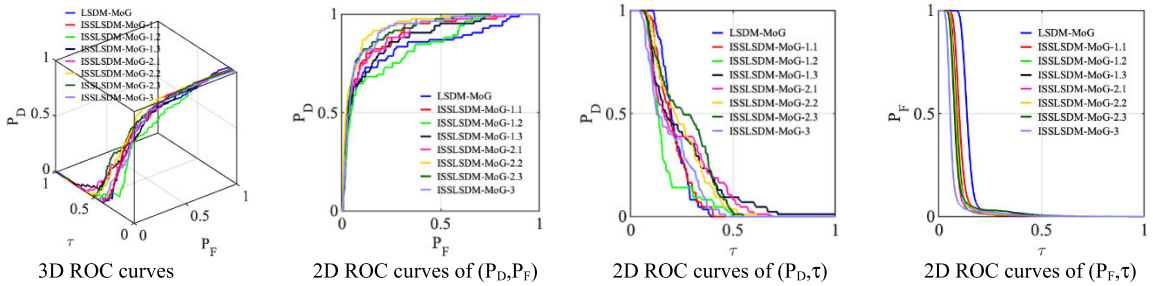
3D ROC curves | 2D ROC curves of (P_D,P_F) | 2D ROC curves of (P_D,τ) | 2D ROC curves of (P_F,τ)

Fig. 44. 3-D ROC curves and the corresponding 2-D ROC curves produced by LRR and its seven ISSLRR variants on the AVIRIS San Diego airport scene.



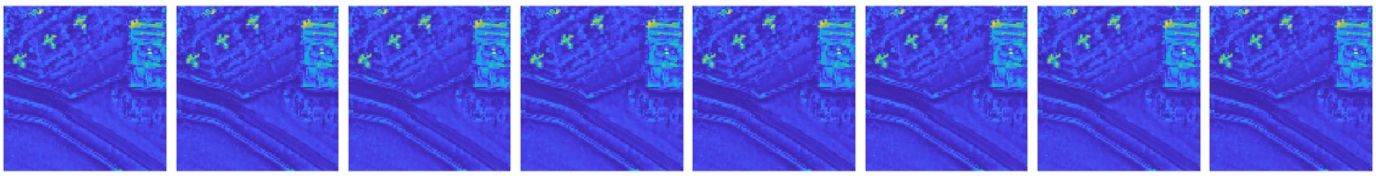
LSDM-MoG | ISSLSDM-MoG -1.1 | ISSLSDM-MoG -1.2 | ISSLSDM-MoG -1.3 | ISSLSDM-MoG -2.1 | ISSLSDM-MoG -2.2 | ISSLSDM-MoG -2.3 | ISSLSDM-MoG -3

Fig. 45. ADMaps produced by LSDM-MoG along with its seven ISS-AD variants on the San Diego airport scene.



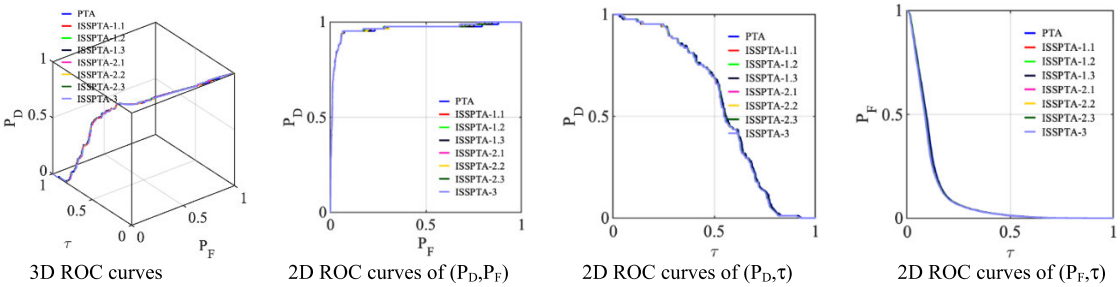
3D ROC curves | 2D ROC curves of (P_D,P_F) | 2D ROC curves of (P_D,τ) | 2D ROC curves of (P_F,τ)

Fig. 46. 3-D ROC curves and the corresponding 2-D ROC curves produced by LSDM-MoG and its seven ISSLSDM-MoG variants on the AVIRIS San Diego airport scene.



PTA | ISSPTA-1.1 | ISSPTA-1.2 | ISSPTA-1.3 | ISSPTA-2.1 | ISSPTA-2.2 | ISSPTA-2.3 | ISSPTA-3

Fig. 47. ADMaps produced by PTA along with its seven ISS-AD variants on the San Diego airport scene.



3D ROC curves | 2D ROC curves of (P_D,P_F) | 2D ROC curves of (P_D,τ) | 2D ROC curves of (P_F,τ)

Fig. 48. 3-D ROC curves and the corresponding 2-D ROC curves produced by PTA and its seven ISSPTA variants on the AVIRIS San Diego airport scene.

ISSLSDM-MoG-3, which had four or three highest AUC values out of eight detection measures.

5) PTA: Fig. 47 shows the ADMaps of PTA along with its corresponding seven ISSPTA variants, ISSPTA-1.1, ISSPTA-1.2, ISSPTA-1.3, ISSPTA-2.1, ISSPTA-2.2, ISSPTA-2.3, and ISSPTA-3 for the San Diego airport scene, where PTA and its ISSPTA also performed significantly better than RXAD in Fig. 39, CRD in Fig. 41, LRR in Fig. 43, and LSDM-MoG in Fig. 45 by visual inspection with three airplanes clearly detected in Fig. 47. However, it is also very obvious to see that PTA performed BKG suppression very poorly. We then plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 48. Table XX calculated the eight detection measures with their AUC values and also

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

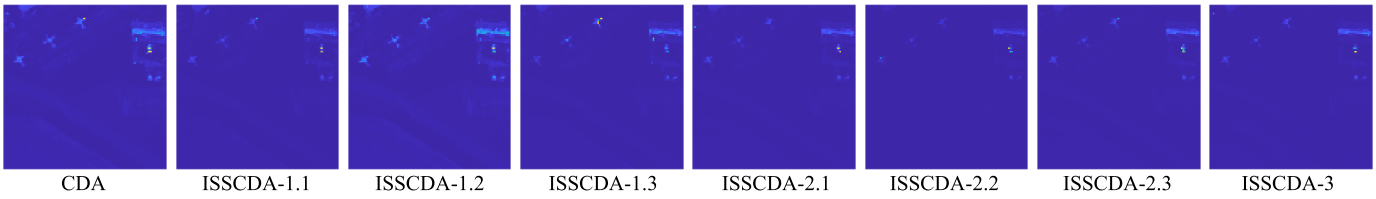


Fig. 49. ADMaps produced by RXAD using CDA along with its seven ISS-AD variants on the San Diego airport scene.

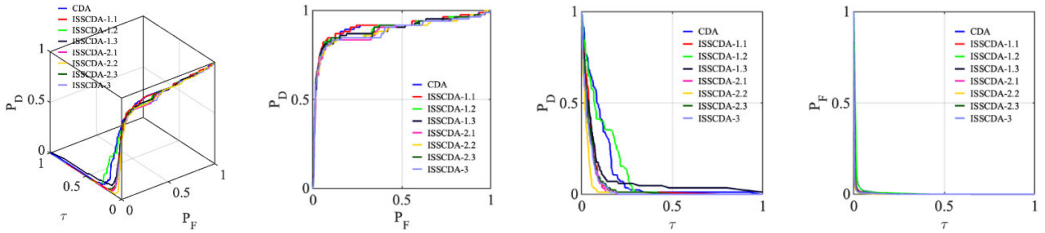


Fig. 50. 3-D ROC curves and the corresponding 2-D ROC curves produced by RXAD using CDA and seven ISSCDA variants on the AVIRIS San Diego airport scene.

TABLE XX
AUC VALUES PRODUCED BY PTA AND ITS SEVEN ISSPTA VARIANTS ON THE AVIRIS SAN DIEGO AIRPORT SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
PTA	0	0.000	0.9613	**0.5501**	0.8883	1.5113	1.8495	0.4383	4.9229	1.4383	104.116
ISSPTA-1.1	1	0.983	0.9632	0.5461	0.8916	1.5092	1.8547	0.4376	5.0362	1.4376	213.760
ISSPTA-1.2	1	0.986	0.9637	0.5479	0.8904	**1.5115**	1.854	0.4382	4.9965	1.4382	215.421
ISSPTA-1.3	1	0.976	0.9624	0.5483	0.8901	1.5106	1.8524	0.4383	4.9866	1.4383	213.681
ISSPTA-2.1	1	0.971	0.9647	0.5442	0.8939	1.5089	1.8586	0.4381	5.1288	1.4381	216.657
ISSPTA-2.2	1	0.971	0.9637	0.5448	0.893	1.5085	1.8567	0.4378	5.0904	1.4378	221.905
ISSPTA-2.3	1	0.975	0.9641	0.5464	0.8921	1.5105	1.8561	**0.4385**	5.0621	**1.4385**	214.648
ISSPTA-3	1	0.957	**0.9650**	0.5434	**0.8949**	1.5084	**1.8600**	0.4384	**5.1724**	1.4384	217.334

TABLE XXI
AUC VALUES PRODUCED BY RXAD USING CDA AND SEVEN ISSCDA VARIANTS ON THE AVIRIS SAN DIEGO AIRPORT SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
CDA	0	0.000	0.907	0.1052	0.9882	**1.0122**	1.8952	0.0934	8.9485	1.0934	0.953
ISSCDA-1.1	9	1.000	**0.9096**	0.0503	0.9941	0.9599	**1.9037**	0.0444	8.5328	1.0444	42.228
ISSCDA-1.2	20	0.651	0.8954	**0.1094**	0.9882	1.0048	1.8836	**0.0976**	9.2714	**1.0976**	90.753
ISSCDA-1.3	20	0.393	0.8924	0.0705	0.994	0.9629	1.8864	0.0645	**11.7798**	1.0645	88.217
ISSCDA-2.1	20	0.025	0.886	0.0391	0.9951	0.9251	1.8811	0.0341	7.94	1.0341	88.097
ISSCDA-2.2	20	0.400	0.8871	0.0262	**0.9971**	0.9133	1.8842	0.0233	8.9645	1.0233	93.489
ISSCDA-2.3	20	0.899	0.8933	0.0448	0.9951	0.938	1.8884	0.0398	9.0887	1.0398	88.298
ISSCDA-3	20	0.817	0.885	0.0372	0.9958	0.9222	1.8808	0.033	8.8884	1.033	88.039

documented the computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original PTA without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original version. As expected, PTA and its ISSPTA variants significantly improved the detection of airplanes over ISSRXAD, ISSCRD, ISSLRR, and ISSLSDM-MoG, with detection rates of three airplanes, and AUC_{ADP} over 0.5 at the expense of poor performance in AUC_{ADBS} and AUC_{BDP}. Interestingly, PTA without ISS feedback produced the highest AUC_{ADP} value but not all the other seven detection measures. Overall speaking, the best one was ISSPTA-3, which had the four highest AUC values out of the eight detection measures.

6) CDA: Fig. 49 shows the ADMaps produced by CDA using RXAD along with its corresponding seven ISSCDA variants, ISSCDA-1.1, ISSCDA-1.2, ISSCDA-1.3, ISSCDA-2.1, ISSCDA-2.2, ISSCDA-2.3, and ISSCDA-3 for the San Diego airport scene, where CDA and its ISSCDA detected three airplanes poorly but did perform very well in BS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

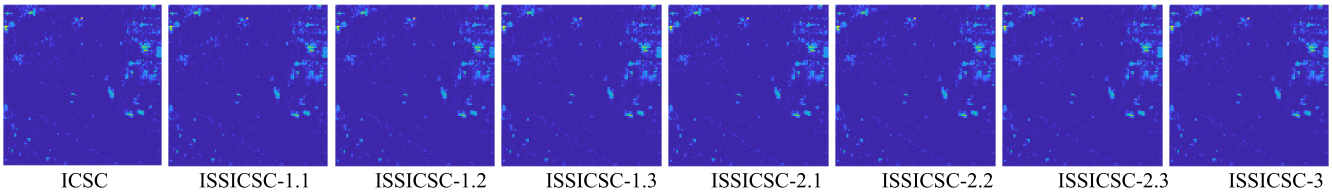


Fig. 51. ADMaps produced by RXAD using ICSC along with its seven ISS-AD variants on the San Diego airport scene.

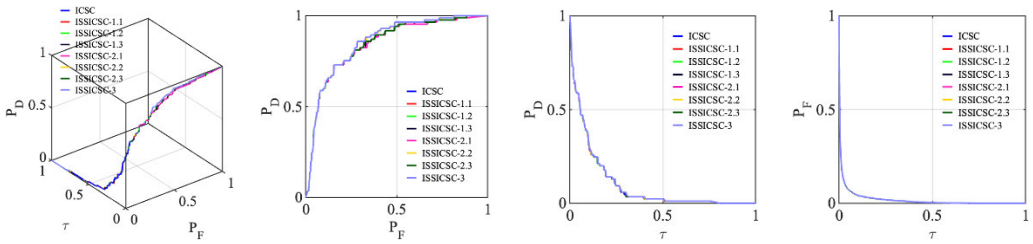


Fig. 52. 3-D ROC curves and the corresponding 2-D ROC curves produced by RXAD using ICSC and its seven ISSICSC variants on the AVIRIS San Diego airport scene.

TABLE XXII
AUC VALUES PRODUCED BY RXAD USING ICSC AND ITS SEVEN ISSICSC VARIANTS ON THE AVIRIS SAN DIEGO AIRPORT SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	Time(s)
ICSC	0	0.000	0.844	0.0976	**0.9808**	0.9415	1.8247	0.0783	5.0679	1.0783	0.479
ISSICSC-1.1	1	0.997	0.844	0.0981	0.9807	0.942	1.8246	0.0787	5.0704	1.0787	8.393
ISSICSC-1.2	1	0.990	0.844	0.0981	0.9807	0.9421	1.8246	0.0788	5.0716	1.0788	8.448
ISSICSC-1.3	1	0.993	0.8464	0.0981	0.9807	0.9445	1.8271	0.0788	5.0723	1.0788	8.305
ISSICSC-2.1	1	0.977	0.844	0.0987	0.9806	0.9427	1.8246	0.0792	5.079	1.0792	8.204
ISSICSC-2.2	1	0.993	0.8561	0.0987	0.9806	0.9548	1.8367	0.0793	5.0816	1.0793	8.271
ISSICSC-2.3	1	0.987	0.8477	0.0987	0.9806	0.9464	1.8282	0.0793	5.0774	1.0793	8.289
ISSICSC-3	1	0.974	**0.8565**	**0.0993**	0.9805	**0.9558**	**1.837**	**0.0798**	**5.0873**	**1.0798**	8.389

We then plotted their 3-D3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 50. Table XXI calculated the eight detection measures with their AUC values and also documented the computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original CDA without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original version. As we can see from Table XXI, CDA and its ISSCDA variants had most of their detection rates of three airplanes, and AUC_{ADP} and AUC_{ADBS} were lower than 0.1 but had very high values in AUC_{BDP}, AUC_{JBS}, and AUC_{SNPR}. Unlike the previous five AD methods, RXAD, CRD, LRR, LSDM-MoG, and PTA, there was no clear winner among CDA and its seven ISSCDA variants. This indicated that CDA itself already performed well, and its improvement by ISS feedback fluctuated.

7) ICSC: Fig. 51 shows the ADMaps produced by ICSC using RXAD along with its corresponding seven ISSICSC variants: ISSICSC-1.1, ISSICSC-1.2, ISSICSC-1.3, ISSICSC-2.1, ISSICSC-2.2, ISSICSC-2.3, and ISSICSC-3 for the San Diego airport scene. Like CDA, ICSC also performed poorly in the detection of three airplanes but did manage to do very well in BS. We then plotted their 3-D ROC curves and generated their corresponding three 2-D ROC curves in Fig. 52. Table XXII calculated the eight detection measures with their AUC values and also documented the computing times and the required numbers of iterations, j, and $TI_{j,j-1}$ values, where the results highlighted by red performed better than original ICSC without ISS feedbacks, and the boldfaced results indicate the best results among all the variants, including the original version. Compared to the CDA results in Table XXI, the ICSC results in Table XXII were slightly worse in every category in eight detection measures with AUC_{ADP} and AUC_{JAD} approximately 0.1 but had very high vales with AUC_{BDP} of 0.098, AUC_{JBS} of 1.82, and AUC_{SNPR} of 5. While there was no best ISSCDA, ISSICSC-3 was a clear winner with the best values across the board. Also, compared to CDA with fluctuated performance, ISSICSC was very stable and consistent.

D. Discussion

This section summarizes the results and conducts a comparative analysis among seven ISSHAD methods based on experiments performed on three data scenes, each of which has its own merit, as discussed in the following.

1) Dataset Analysis: The HYDICE panel scene offers great advantages that the other two cannot. It contains 19 panel pixels that can be considered as anomalous targets specified by three types of data samples, two-pixel pure pixels in the first column, one-pixel mixed sample in the second column,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

and subpixels in the third column. This scene allows one to test and evaluate the effectiveness of an anomaly detector in detecting pure, mixed, and subpixel anomalies. It is a versatile dataset and can be used for a wide variety of applications. The HYDICE urban scene contains only 21 anomalous pixels that can be considered anomalies. It is particularly used for AD since the anomalies are randomly scattered and no particular spatial correlation surrounding the anomalies. The San Diego airport scene contains three airplanes that are too large to be considered anomalies. This scene is suitable for target detection not AD, as pointed out in [27]. As expected, a detector, which works well for the HYDICE urban scene as an anomaly detector, may not work well for the vSan Diego airport scene as a target detector, such as RXAD and CRD. Conversely, a detector, which works well for the San Diego airport scene as a target detector, may also not work well on the HYDICE scene, such as PTA. As a consequence, conclusions drawn from one scene cannot be also applied to the other scene. This fact was confirmed by experiments performed in Sections VI-B and VI-C.

2) Comparative Analysis Among Test Anomaly Detectors:

(i) The first and most important conclusion is that ISSHAD indeed improved significantly RXAD, CRD, LRR, LSDM-MoG, and PTA in AD, which did not use ISS feedback in all three data scenes.

(ii) Among the three types of spectral and spatial information used as feedback, the SFMap is crucial because it includes spatial information surrounding anomalies. This is evidenced by the fact that most effective ISSHAD variants include SFMaps as their feedback.

(iii) The reason that ISSHAD could not improve CDA is due to the fact that CDA could not remove Gaussian-like BKG and interference in its PC space. As a result, such leakage was magnified by feedback. This was evidenced that a large number of iterations with low TI values were required to terminate CDA. However, ICSC did not have such an instability issue because it used DS to remove all potential interferences characterized by second-order statistics.

(iv) RXAD using ICSC performed better than RXAD using CDA for the HYDICE panel data scene, while RXAD using CDA performed slightly better than RXAD using ICSC for the HYDICE urban data scene and the San Diego airport data scene at the expense of its instability caused by unstable iterations. Thus, overall speaking, the best among all test AD methods was the RXAD using the ICSC space, which also confirmed the conclusions in [27].

(v) Generally speaking, the best among all the seven ISSHAD variants was ISSHAD-3 followed by ISSHAD-2.3. This indicated that feeding back all three maps, ADMap, FGMap, and SFMap, gave the best performance. In particular, in terms of providing better anomaly information, it seemed that the SFMap is the best followed by FGMap and then by ADmap. Since anomalies are closely related to their neighboring data sample vectors, SFMap offers its needed information. On the other hand, despite that FGMap does not include spatial information as SFMap does, it only retains the information of detected anomalies without BKG. Thus, feeding back FGMap can enhance the spectral information of anomalies. As for ADMap, it contains the information on all detected anomalies and BKG. If an anomaly detector works effectively, the feedback information will contain the information of enhanced anomalies with suppressed BKG. The resulting ISSHAD definitely improves AD without ISS feedback. Nevertheless, the general guideline for implementing ISSHAD is to use ISSHAD-3.

(vi) It should be noted that many recently developed model- or DL network-based AD methods have been focused on how to remove and suppress BKG, and have not paid much attention to designing anomaly detectors. For example, LRR, CDA, and ICSC use RXAD to perform AD where RXAD is the Mahalanobis distance as an l_2-norm weighted by inverting covariance matrix $\mathbf{K}$. As for LSDM-MoG, it uses the Manhattan distance that can be considered as l_1-norm. In regard to CRD and PTA, both use the Euclidean distance that can be considered as an unweighted l_2-norm to perform AD in the residual space and anomaly tensor space, respectively.

(vii) Despite that ISSHAD can improve an AD method, on some occasions, it will run into an issue that the used AD method results in fluctuations such as CDA due to unknown random signal sources. To resolve this issue, we can add an additional stopping rule by comparing the result from the first iteration. If the result obtained from the first iteration is worse than the result without feedback, the ISSHAD is terminated, in which case, no feedback is necessary.

(viii) Since model-based AD methods, such as LRR and LSDM-MoG, generally require solving optimization problems repeatedly, as a result, their time complexity is usually very high and requires more iterative feedback than those without using models such as RXAD and CRD

(ix) The time complexity of ISSHAD is determined by the number of iterations required to terminate ISSHAD. Thus, it is more appropriate to evaluate the AD methods without using iterations. In this case, the best one was RXAD followed by ICSC, CDA, CRD, LSDM-MoG, and PTA with the worst LRR where ICSC and CDA had comparable times with slight differences followed by CRD and LSDM-MoG with time differences by small scale and concluded by PTA and LRR with excessive time complexity. Thus, with all things considered, we can conclude that the best one is RXAD using ICSC followed by the second best RXAD using CDA and the worst ones were PTA and LRR.

(x) Last but not least, the San Diego airport scene experiments also confirmed the findings in [27], which showed that the San Diego airport scene was not good for AD. As shown in our experiments, all test AD methods except PTA barely detected airplanes with very

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

low AUC_{ADP} values from 0.08 by RXAD to 0.21 by LSDM-MoG. This means that airplanes were indeed not considered anomalies by these AD methods. Nevertheless, their $AUC_{(D,F)}$ values were still very high over 0.83. On the other hand, PTA produced the highest $AUC_{(D,F)} = 0.96$ and the highest $AUC_{ADP} = 0.55$. Thus, if the AD performance is evaluated solely based on $AUC_{(D,F)}$, PTA was the best among the test AD methods. Unfortunately, Fig. 47 clearly showed that PTA had the worst BS. These studies further demonstrated that conclusions drawn from the San Diego scene are generally misleading.

VII. Conclusion

This article presents a new approach to AD, ISSHAD, which can increase and enhance an anomaly detector in its detection performance. It introduces an iterative process into an anomaly detector to iteratively include spectral and spatial information obtained from its produced ADMaps in previous iterations. In order to obtain spectral information about detected anomalies, three types of spectral anomaly maps are fed back. One is ADMaps directly produced by the anomaly detector. Another is the spectral FG anomaly map obtained from Otsu's thresholded BMap by retaining the FG data samples while setting all the BKG data samples to zeros. A third one is spatial SFMap that includes spatial information around detected anomalies via an SF. The level of improvement is determined by feedback on these three types of information. Combining different feedbacks of spectral and spatial information gives rise to seven variants of ISSHAD. To illustrate and demonstrate how IHSSAD can effectively improve an anomaly detector, the test anomaly detectors are selected from the most widely used and several recently developed AD methods, RXAD, CRD, LRR, LSDM-MoG, PTA, CDA, and ICSC. The conducted experiments showed that their corresponding ISSHAD counterparts indeed performed better than their original versions. If the used anomaly already performs very well, the improvement by ISSHAD may be limited. If there is no further improvement, ISSHAD will be terminated at the first iteration. If an anomaly detector performs not well, the improvement by ISSHAD can be very significant.

References

[1] C.-I Chang, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification*. Norwell, MA, USA: Kluwer Academic/Plenum, 2003.

[2] C.-I Chang, *Hyperspectral Data Processing: Algorithm Design and Analysis*. Hoboken, NJ, USA: Wiley, 2013.

[3] C.-I Chang, *Real-Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection*. New York, NY, USA: Springer, 2016.

[4] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[5] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–218, Jan. 1992.

[6] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Trans. Signal Process.*, vol. 42, no. 8, pp. 2146–2157, Aug. 1994.

[7] S. Kraut and L. L. Scharf, "The CFAR adaptive subspace detector is a scale-invariant GLRT," *IEEE Trans. Signal Process.*, vol. 47, no. 9, pp. 2538–2541, Sep. 1999.

[8] S. Kraut, L. Scharf, and L. T. McWhorter, "Adaptive subspace detectors," *IEEE Trans. Signal Process.*, vol. 49, no. 1, pp. 1–16, Jan. 2001.

[9] D. W. J. Stein, S. G. Beaven, L. E. Hoff, E. M. Winter, A. P. Schaum, and A. D. Stocker, "Anomaly detection from hyperspectral imagery," *IEEE Signal Process. Mag.*, vol 19, no. 1, pp. 58–69, Jan. 2002.

[10] D. Manolakis, E. Truslow, M. Pieper, T. Cooley, and M. Brueggeman, "Detection algorithms in hyperspectral imaging systems," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 24–33, Jan. 2014.

[11] C.-I Chang, "Hyperspectral target detection: Hypothesis testing, signal-to-noise ratio, and spectral angle theories," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505223.

[12] C.-I Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511720.

[13] C.-I Chang, "Constrained energy minimization anomaly detection for hyperspectral imagery via dummy variable trick," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517119.

[14] C.-I Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.

[15] C.-I Chang, K. Y. Ma, C. C. Liang, Y. M. Kuo, and S. Chen, "Iterative random training sampling spectral-spatial classification for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3986–4007, 2020.

[16] S. Zhong, C.-I Chang, and Y. Zhang, "Iterative edge preserving filtering approach to hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 90–94, Jan. 2019.

[17] S. Zhong, C.-I Chang, and Y. Zhang, "Spectral–spatial feedback close network system for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10056–10069, Dec. 2019.

[18] C.-I Chang, "Band sampling for hyperspectral imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514024.

[19] C.-I Chang and K. Y. Ma, "Band sampling of kernel constrained energy minimization using training samples for hyperspectral mixed pixel classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522721.

[20] N. Otsu, "A threshold selection method from gray-level histgram," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.

[21] S. Theodoridis and K. Koutroumbas, *Pattern Recognition*. Cambridge, MA, USA: Academic Press, 1999, p. 366.

[22] L. Wei and D. Qian, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[23] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2015.

[24] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[25] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[26] S. Chen, C.-I Chang, and X. Li, "Component decomposition analysis for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516222.

[27] C.-I Chang, "Effective anomaly space for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526624.

[28] C.-I Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.

[29] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[30] S. Matteoli, T. Veracini, M. Diani, and G. Corsini, "A locally adaptive background density estimator: An evolution for RX-based anomaly detectors," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 323–327, Jan. 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.

[31] A. Banerjee, P. Burlina, and C. Diehl, "A support vector method for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2282–2291, Aug. 2006.

[32] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[33] C.-I Chang, Y. Wang, and S.-Y. Chen, "Anomaly detection using causal sliding windows," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 7, pp. 3260–3270, Jul. 2015.

[34] H. Kwon, S. Z. Der, and N. M. Nasrabadi, "Adaptive anomaly detection using subspace separation for hyperspectral imagery," *Opt. Eng.*, vol. 42, no. 11, pp. 3342–3351, 2003.

[35] W. Liu and C.-I Chang, "Multiple window anomaly detection for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 658–664, Apr. 2013.

[36] W. Xie, T. Jiang, Y. Li, X. Jia, and J. Lei, "Structure tensor and guided filtering-based algorithm for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4218–4230, Jul. 2019.

[37] J. Lei, W. Xie, J. Yang, Y. Li, and C.-I Chang, "Spectral–spatial feature extraction for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8131–8143, Oct. 2019.

[38] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[39] Y. Wang et al., "A posteriori hyperspectral anomaly detection for unlabeled classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3091–3106, Jun. 2018.

[40] S. Y. Chen, Y. Wang, C. C. Wu, C. Liu, and C.-I Chang, "Real time causal processing of anomaly detection in hyperspectral imagery," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 50, no. 2, pp. 1511–1534, Apr. 2014.

[41] J. Liu, Z. Hou, W. Li, R. Tao, D. Orlando, and H. Li, "Multipixel anomaly detection with unknown patterns for hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5557–5567, Oct. 2022.

[42] Y. Yuan, D. Ma, and Q. Wang, "Hyperspectral anomaly detection by graph pixel selection," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 3123–3134, Dec. 2016.

[43] S. Li, K. Zhang, Q. Hao, P. Duan, and X. Kang, "Hyperspectral anomaly detection with multiscale attribute and edge-preserving filters," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1605–1609, Oct. 2018.

[44] N. Ma, Y. Peng, S. Wang, and P. Leong, "An unsupervised deep hyperspectral anomaly detector," *Sensors*, vol. 18, no. 3, p. 693, Feb. 2018, doi: 10.3390/s18030693.

[45] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Deep low-rank prior for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527017.

[46] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.

[47] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.

[48] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020.

[49] J. Qu, Q. Du, Y. Li, L. Tian, and H. Xia, "Anomaly detection in hyperspectral imagery based on Gaussian mixture model," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9504–9517, Nov. 2021.

[50] R. Zhao, B. Du, and L. Zhang, "A robust nonlinear hyperspectral anomaly detection approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1227–1234, Apr. 2014.

[51] R. Zhao, B. Du, and L. Zhang, "Hyperspectral anomaly detection via a sparsity score estimation framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3208–3222, Jun. 2017.

[52] S. Chang, B. Du, and L. Zhang, "BASO: A background-anomaly component projection and separation optimized filter for anomaly detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3747–3761, Jul. 2018.

[53] E. J. Candes, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *J. ACM*, vol. 58, no. 3, pp. 1027–1063, 2009.

[54] N. Vaswani, T. Bouwmans, S. Javed, and P. Narayanamurthy, "Robust subspace learning: Robust PCA, robust subspace tracking, and robust subspace recovery," *IEEE Signal Process. Mag.*, vol. 35, no. 4, pp. 32–55, Jul. 2018.

[55] N. Vaswani, Y. Chi, and T. Bouwmans, "Rethinking PCA for modern data sets: Theory, algorithms, and applications," *Proc. IEEE*, vol. 106, no. 8, pp. 1274–1276, Aug. 2018.

[56] T. Bouwmans, A. Sobral, S. Javed, S. K. Jung, and E.-H. Zahzah, "Decomposition into low-rank plus additive matrices for background/foreground separation: A review for a comparative evaluation with a large-scale dataset," *Comput. Sci. Rev.*, vol. 23, pp. 1–71, Jan. 2017.

[57] A. Sobral, T. Bouwmans, and E. Zahzah, "LRSLibrary: Low-rank and sparse tools for background modeling and subtraction in videos," in *Handbook on Robust Low-Rank and Sparse Matrix Decomposition: Applications in Image and Video Processing*, T. Bouwmans, N. Aybat, and E. Zahzah, Eds. Boca Raton, FL, USA: CRC Press, May 2016.

[58] T. Zhou and D. Tao, "GoDec: Randomized low-rank & sparsity matrix decomposition in noisy case," in *Proc. 28th Int. Conf. Mach. Learn.*, Bellevue, WA, USA, 2011, pp. 1–16.

[59] S. Wang, X. Wang, Y. Zhong, and L. Zhang, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314.

[60] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1389–11376, Mar. 2016.

[61] B. Xue et al., "A subpixel target approach to hyperpsectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5093–5114, Sep. 2017.

[62] X. Kang, S. Li, and J. A. Benediktsson, "Spectral–spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.



Chein-I Chang (Life Fellow, IEEE) received the B.S. degree in mathematics from Soochow University, Taipei, Taiwan, in 1973, the M.S. degree in mathematics from the Institute of Mathematics, National Tsing Hua University, Hsinchu, Taiwan, in 1975, the M.A. degree in mathematics from the State University of New York at Stony Brook, Stony Brook, NY, USA, in 1977, the M.S. and M.S.E.E. degrees from the University of Illinois at Urbana–Champaign, Urbana, IL, USA, in 1982, and the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

He has been with the University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, since 1987, where he is currently a Professor with the Department of Computer Science and Electrical Engineering. He is currently a Yushan Scholar Chair Professor with National Cheng Kung University, Tainan, Taiwan. He has been holding a Chang Jiang Scholar Chair Professorship and the Director of the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Dalian Maritime University, Dalian, China, since 2016. He has authored four books: *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic Publishers, 2003), *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection* (Springer, 2016), and *Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer, 2017). He is working on his fifth book on Hyperspectral Target and Anomaly Detection: Statistical Signal and Image Processing Perspectives to be published by Wiley in 2024. In addition, he also edited two books, *Recent Advances in Hyperspectral Signal and Image Processing* (2006) and *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007), and coedited with A. Plaza a book *High Performance Computing in Remote Sensing* (CRC Press, 2007). His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of Society for Photo-Optical Instrumentation Engineers (SPIE). He received the National Research Council Senior Research Associateship Award from 2002 to 2003 sponsored by the U.S. Army Soldier and Biological Chemical Command, Edgewood Chemical and Biological Center, Aberdeen Proving Ground, MD, USA. He is serving as an Associate Editor for *Remote Sensing* and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He has been a guest editor of several special issues.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.



Chien-Yu Lin received the B.S. degree in electrical engineering from National Central University, Taoyuan City, Taiwan, in 2013, and the M.S. degree in electrical engineering from National Cheng Kung University, Tainan, Taiwan, in 2015. She is currently pursuing the Ph.D. degree in electrical engineering with the Remote Sensing Signal and Image Processing Laboratory, University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA.

Her research interests include hyperspectral imaging, remote sensing, medical data analysis, machine learning, and digital signal processing.



Pau-Choo (Julia) Chung (Fellow, IEEE) received the Ph.D. degree in electrical engineering from Texas Tech University, Lubbock, TX, USA, in 1991.

She then joined the Department of Electrical Engineering, National Cheng Kung University (NCKU), Tainan, Taiwan, in 1991, where she has become a Full Professor in 1996. She was the Head of the Department of Electrical Engineering and the Director of the Institute of Computer and Communication Engineering, NCKU, from 2011 to 2014 and 2008 to 2011, respectively. She was an elected Distinguished Professor with NCKU in 2005. She was the Program Director of the Intelligent Computing Division, Ministry of Science and Technology from 2012 to 2014, the Director General of the Department of Information and Technology Education at the Ministry of Education, and Dean of the Miin Wu School of Computing, NCKU. She is currently the Dean of the College of Electrical Engineering and Computer Science, NCKU. Her research interests include computational intelligence, machine learning, computational pathology, medical image analysis, and pattern recognition. She is also focusing on whole slide image (WSI) pathology image analysis and has built the ALOVAS platform.

Dr. Chung has served on two terms as an ADCOM Member from 2009 to 2011 and 2012 to 2014. She also served on two terms as a BoG Member of the IEEE Circuit and Systems Society. She is a member of the Phi Tau Phi Honor Society. She received the Distinguished Professor Award of the Chinese Institute of Electrical Engineering in 2012. She has served as the Chair of the CIS Distinguished Lecturer Program from 2012 to 2013 and the Women in Computational Intelligence in 2014 and the Vice-President for Members Activities of the IEEE CIS Society. She has served as an Associate Editor for IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS and IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS. She is also an Associate Editor of IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE. She also serves as the Vice-President for Education of the IEEE CIS Society.



Peter Fuming Hu received the B.S. degree in electrical engineering from Shanghai University, Shanghai, China, in 1984, and the M.S. and Ph.D. degrees in computer science from the University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, in 1992 and 2013, respectively.

He has been with the School of Medicine, UMBC, since 1992, where he is currently an Associate Professor with the Department of Anesthesiology, Surgery and Epidemiology. He also holds the Adjunct Associate Professor at the Department of Computer Science and Electrical Engineering, UMBC. He is also the Chief Technologist with the University of Maryland Shock Trauma Center. He has more than 25 years of experience in clinical research at the Shock Trauma Center, University of Maryland School of Medicine. During that time, he worked as a PI, a Co-PI, or a Co-Investigator on over 50 successfully completed peer-reviewed grants funded by the National Science Foundation (NSF), the U.A. Army, US Air Force (USAF), the Office of Naval Research (ONR), National Aeronautics Space Administration (NASA), National Library of Medicine/National Institute of Health (NLM/NIH), and the Agency for Healthcare Research and Quality (AHRQ). He has published more than 100 peer-reviewed clinical journal articles. The focus of his research is on applying big data machine learning (AI) methods for the prediction of trauma patient outcomes.

Dr. Hu is also a member of the Editorial Board of *Critical Care Medicine* that is the leading clinical journal dedicated to ICU patient care.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:58 UTC from IEEE Xplore. Restrictions apply.